

Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 29, 2026

1 Overview

This document compiles every current SCM case report TeX file under results/.

2 Case 1: cases99 80x2m 9h report

3 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/CASES99/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

3.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/cases99`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

3.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\min, \text{surf}}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

3.3 GSPT Bifurcation & Manifold Calibration

3.4 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399988, 94.108850] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-2.012 \times 10^{-6}, 1552.767352]$

Table 2: GSPT Normal Form Calibration Parameters for idealized_sbl

Metric	Symbol	Estimated Value
Report / Calibration Status	—	invalid_fit
Estimated Fold Richardson Number	$\hat{Ri}_{\text{fold}}$	NA
95% MBB Confidence Interval	CI ₉₅	[NA, NA]
Transcritical Ignition Proxy	$\hat{Ri}_{\text{critical}}$	0.000000
Hysteresis Width	ΔRi_H	NA
Scaling Exponent	$\hat{\gamma}$	NA
Normal Form Amplitude	$\hat{c}$	NA

3.5 Generated Artifacts

- Time-series CSV: `results/cases99/time_series.csv`
- Diagnostic summary JSON: `results/CASES99/summary.json`
- Payload JLD2: `results/cases99/payload.jld2`

3.6 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)
8. Fold proximity diagnostics versus time

3.7 Included Plots

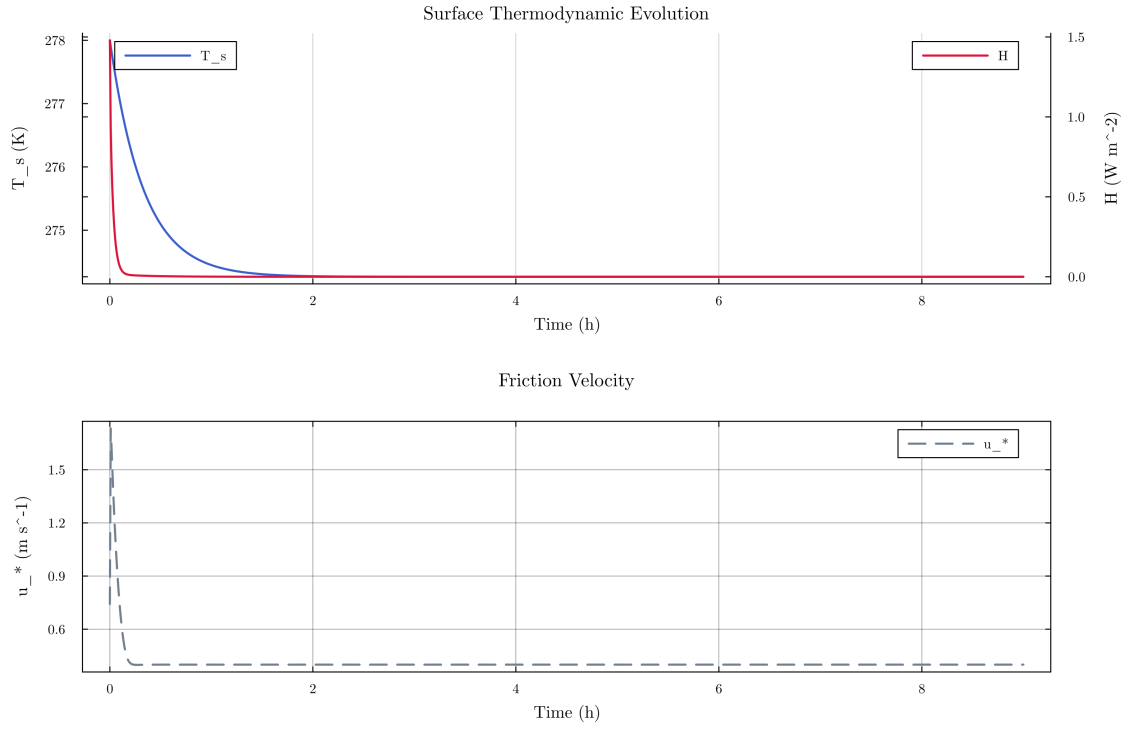


Figure 1: Time series of T_s , sensible heat flux, and friction velocity

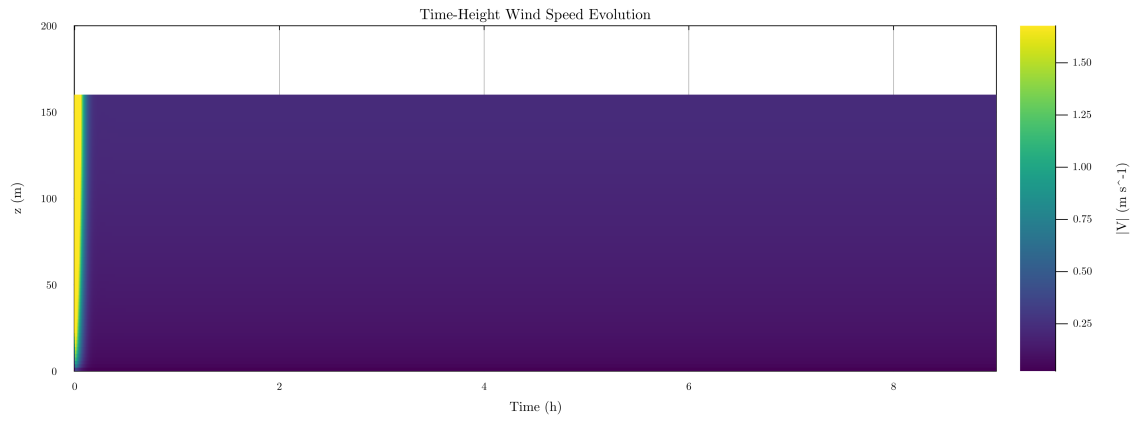


Figure 2: Time-height contour of wind speed with LLJ evolution

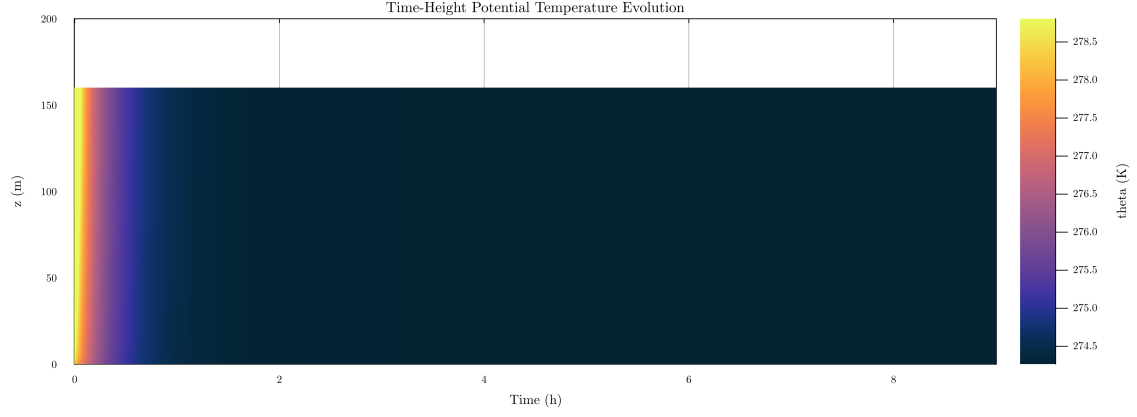


Figure 3: Time-height contour of potential temperature and inversion growth

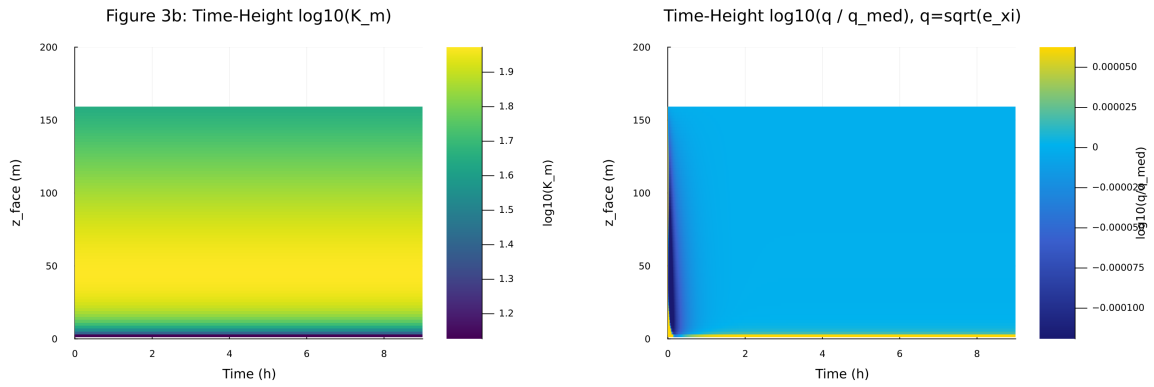


Figure 4: Vertical profiles of U , θ , and K_m at representative times

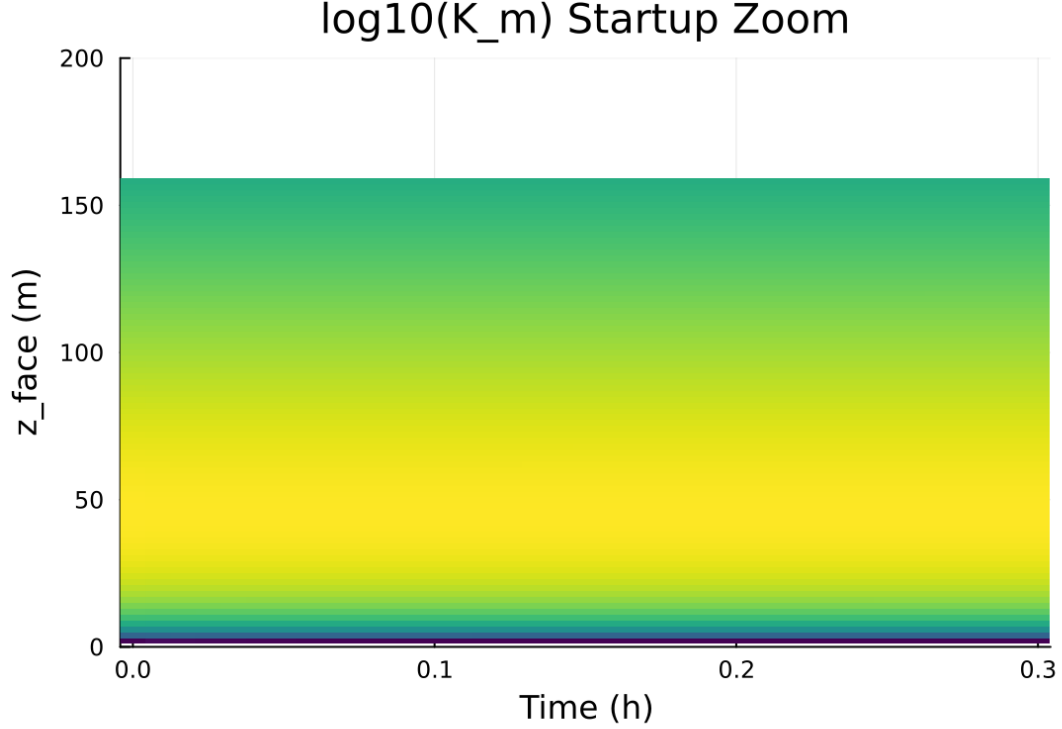


Figure 5: Surface energy budget components (R_n , H , G , storage)

4 Case 2: floss 80x2m 9h report

5 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/FLOSS/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

5.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/floss`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

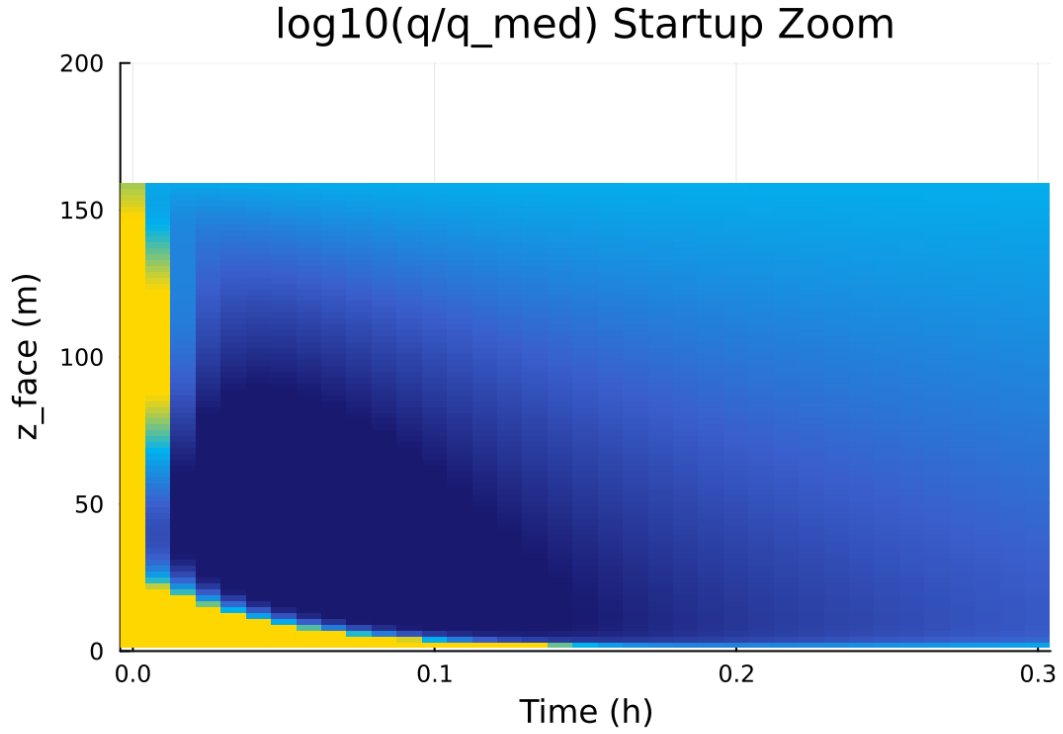


Figure 6: Phase portrait on regularized manifold (Delta vs e_{xi})

Table 3: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

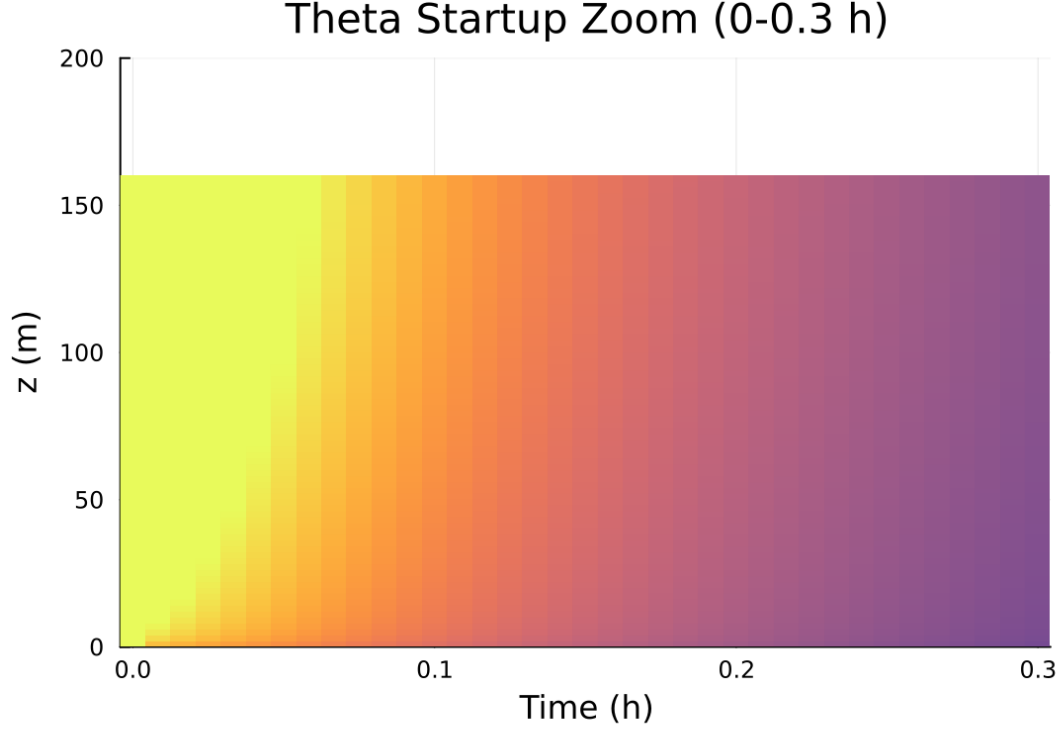


Figure 7: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)

5.2 Forcing and Boundary Conditions

5.3 GSPT Bifurcation & Manifold Calibration

Table 4: GSPT Normal Form Calibration Parameters for idealized_sbl

Metric	Symbol	Estimated Value
Report / Calibration Status	—	invalid_fit
Estimated Fold Richardson Number	$\hat{Ri}_{\text{fold}}$	NA
95% MBB Confidence Interval	CI_{95}	[NA, NA]
Transcritical Ignition Proxy	$\hat{Ri}_{\text{critical}}$	0.000000
Hysteresis Width	ΔRi_H	NA
Scaling Exponent	$\hat{\gamma}$	NA
Normal Form Amplitude	$\hat{c}$	NA

5.4 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399988, 94.108850] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-2.012 \times 10^{-6}, 1552.767352]$

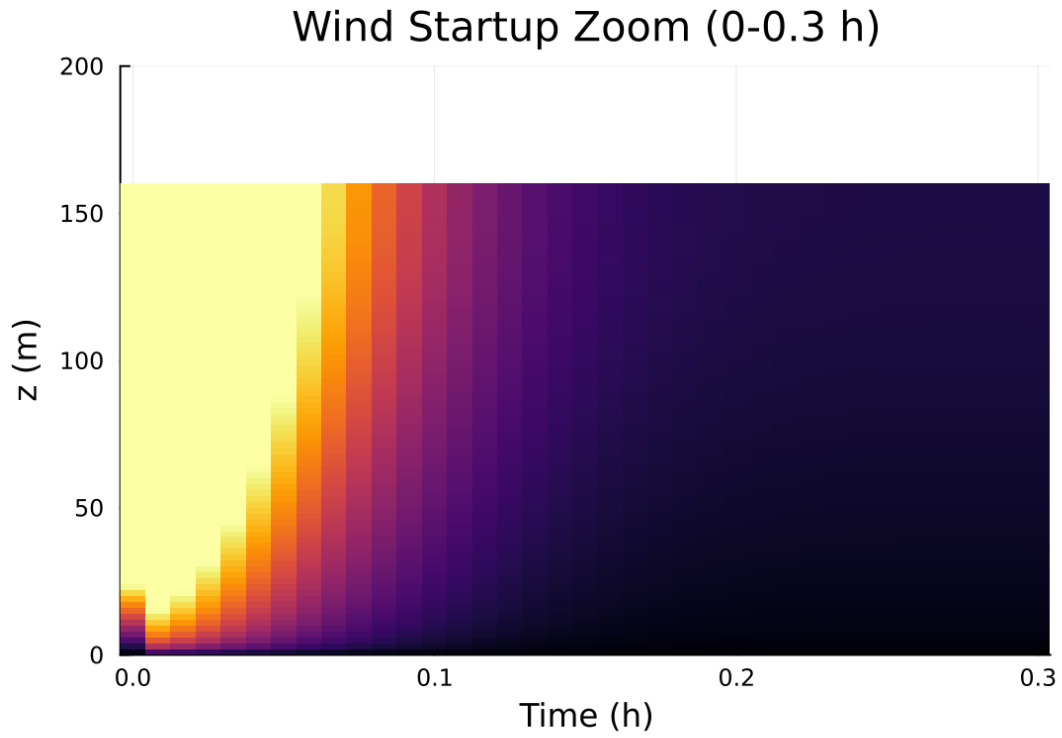


Figure 8: Fold proximity diagnostics versus time

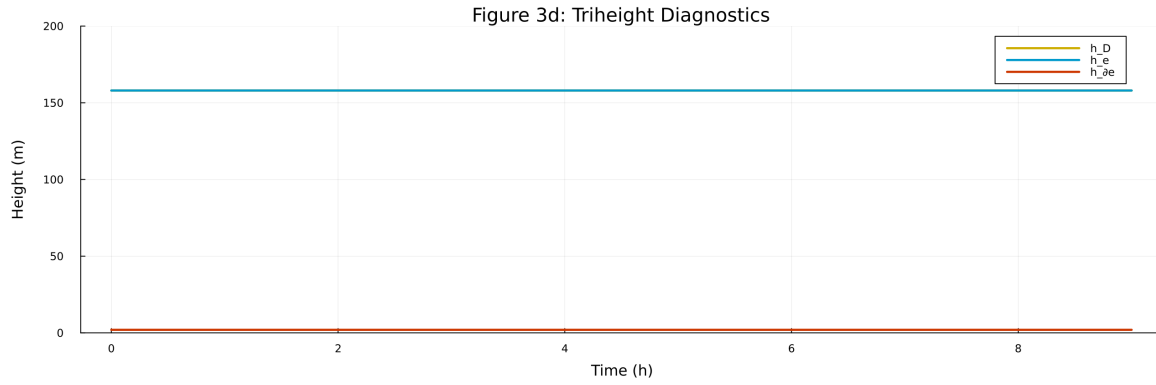


Figure 9: fig03d triheight timeseries.png

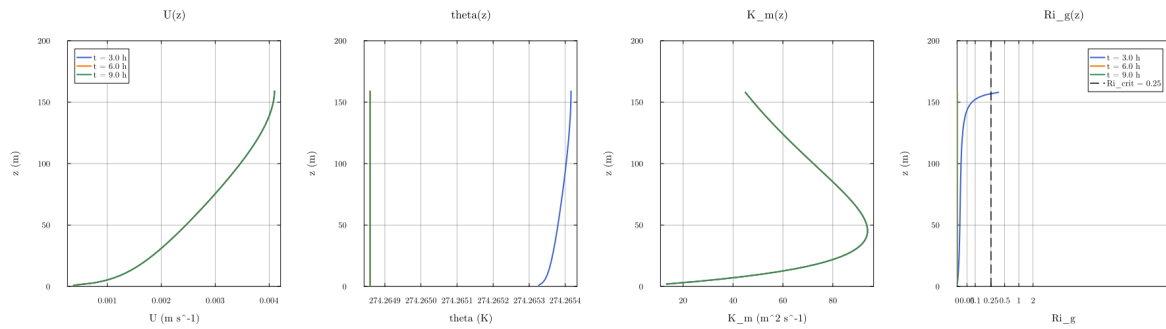


Figure 10: fig04 profiles u theta km.png

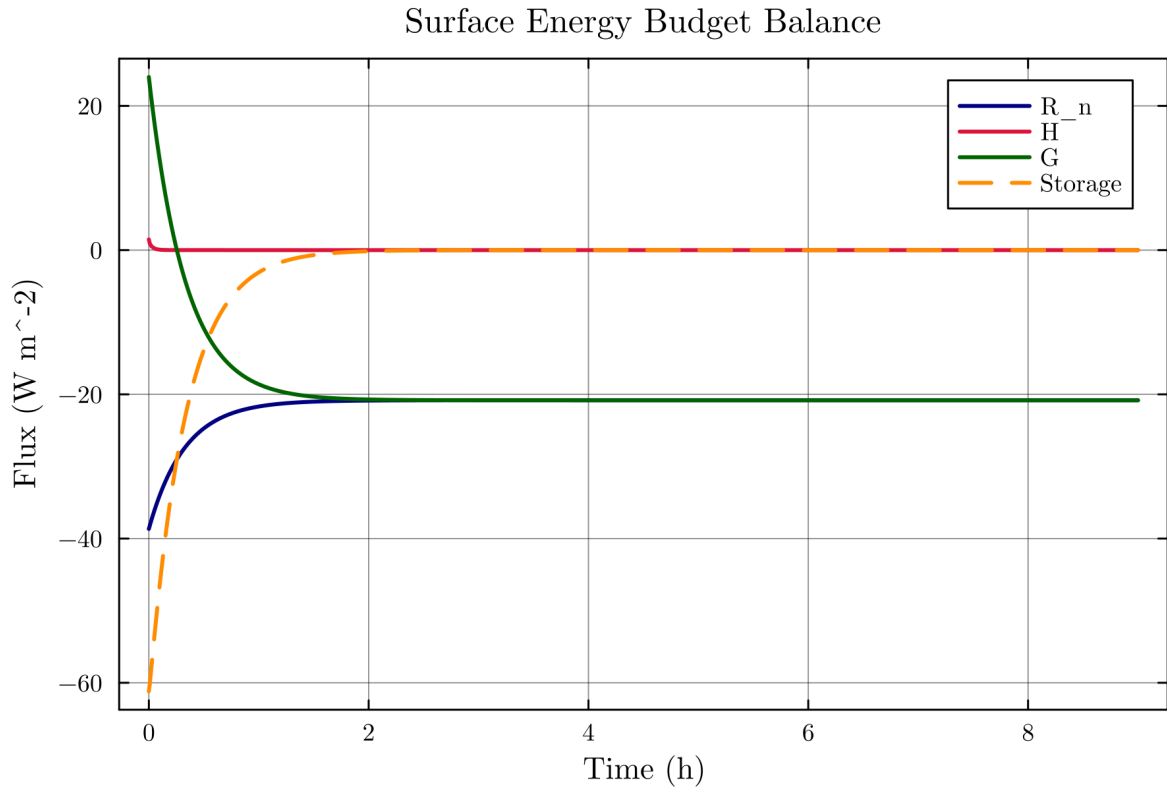


Figure 11: fig05 surface energy budget.png

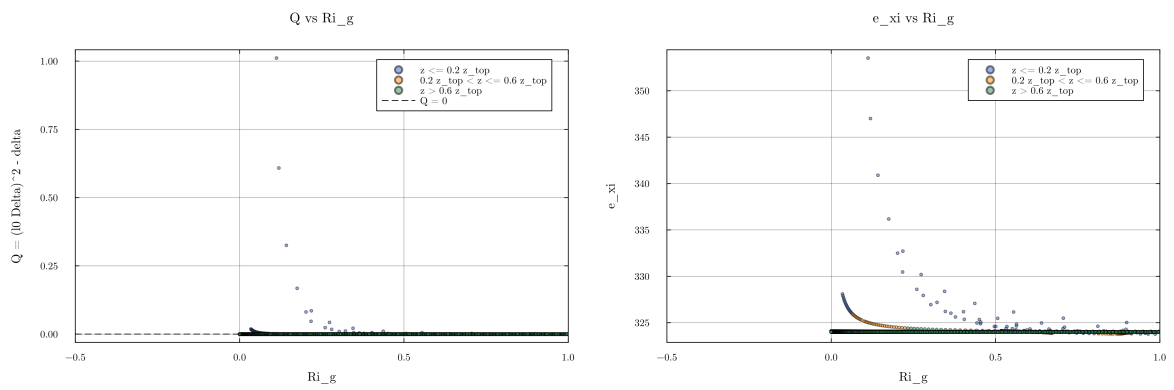


Figure 12: fig06 phase delta exi.png

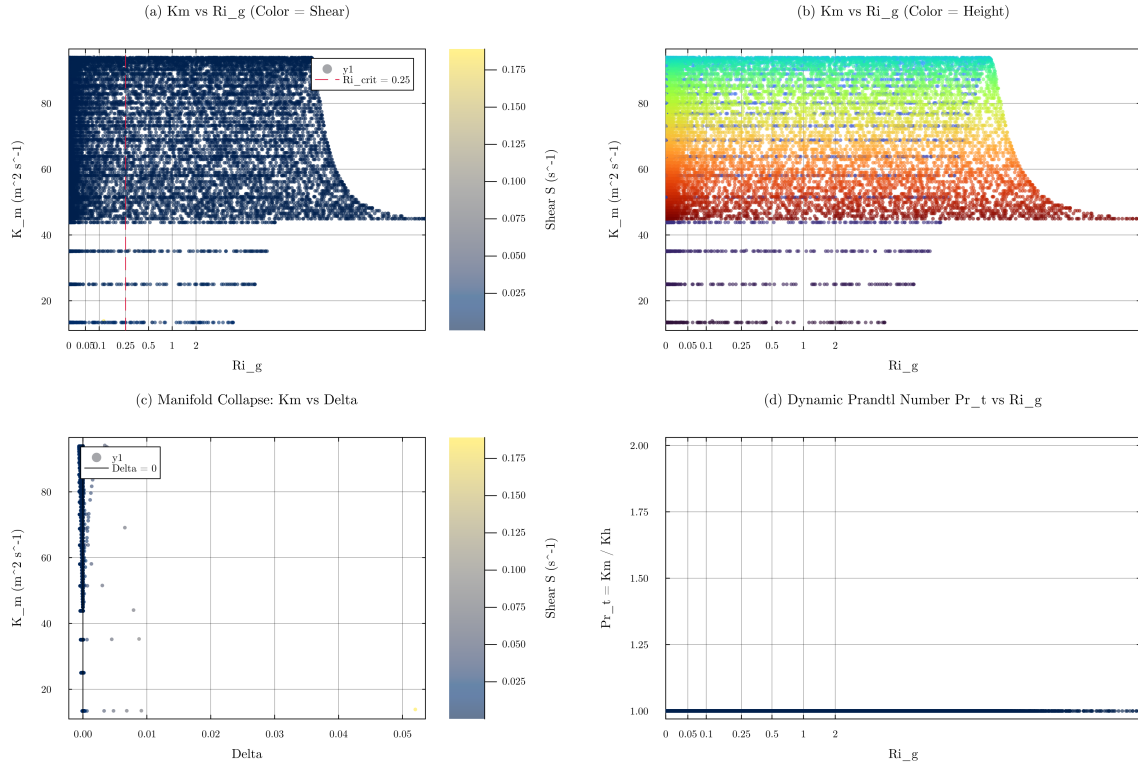


Figure 13: fig07 closure manifold.png

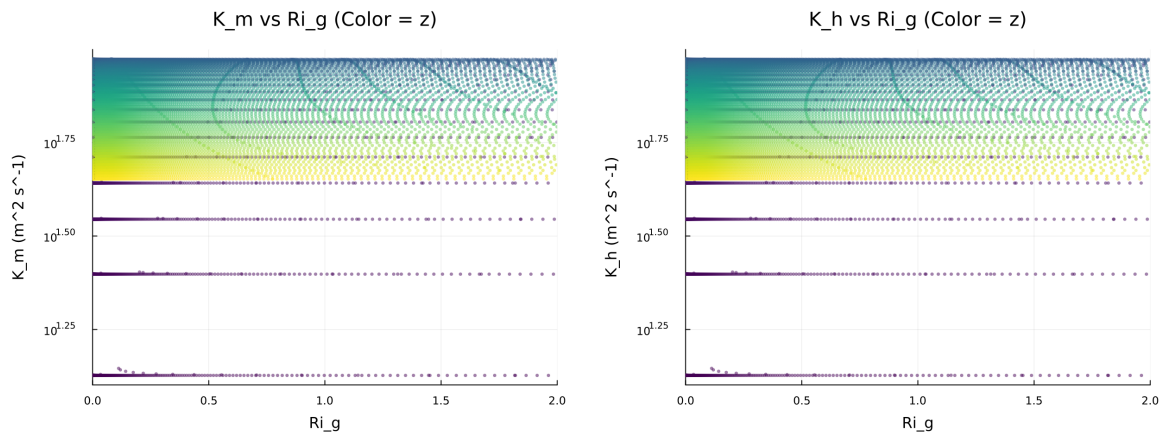


Figure 14: fig07 diffusivity vs ri.png

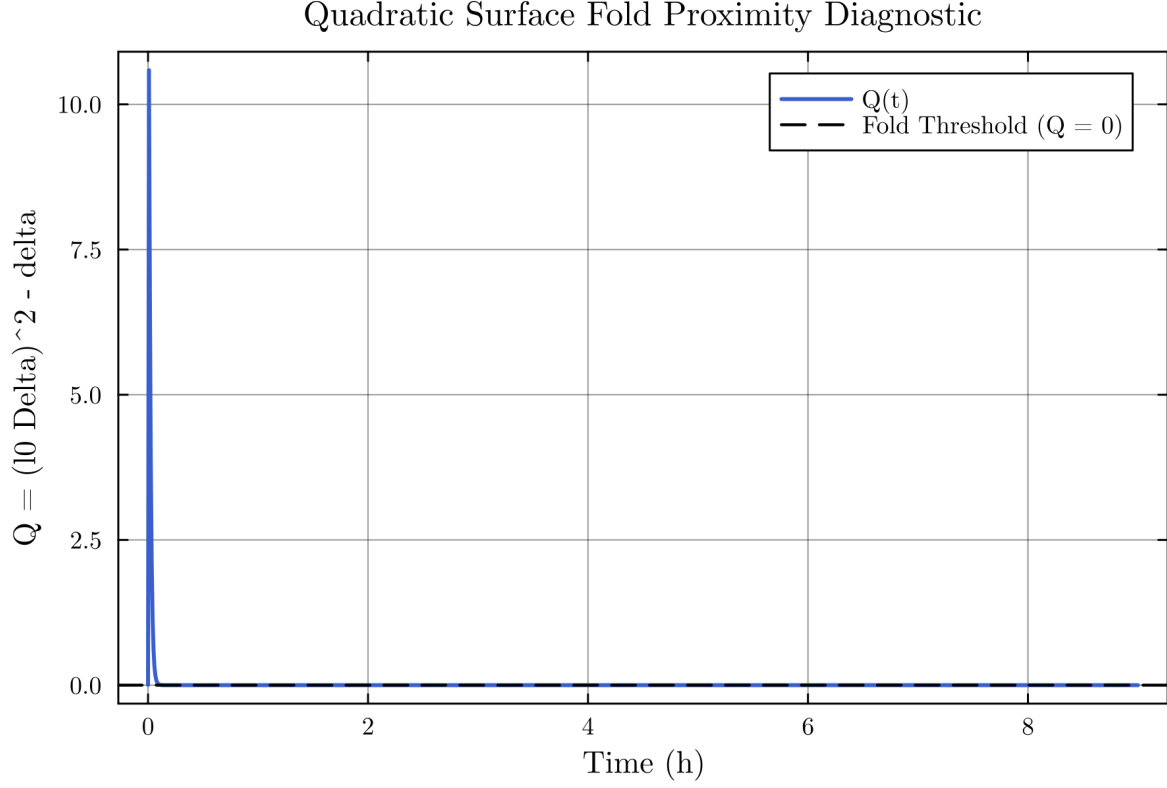


Figure 15: fig08 fold proximity.png

5.5 Generated Artifacts

- Time-series CSV: `results/floss/time_series.csv`
- Diagnostic summary JSON: `results/FLOSS/summary.json`
- Payload JLD2: `results/floss/payload.jld2`

5.6 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Closure manifold diagnostics (K_m vs Ri_g and K_m vs Δ)
8. Fold proximity diagnostics versus time

5.7 Included Plots

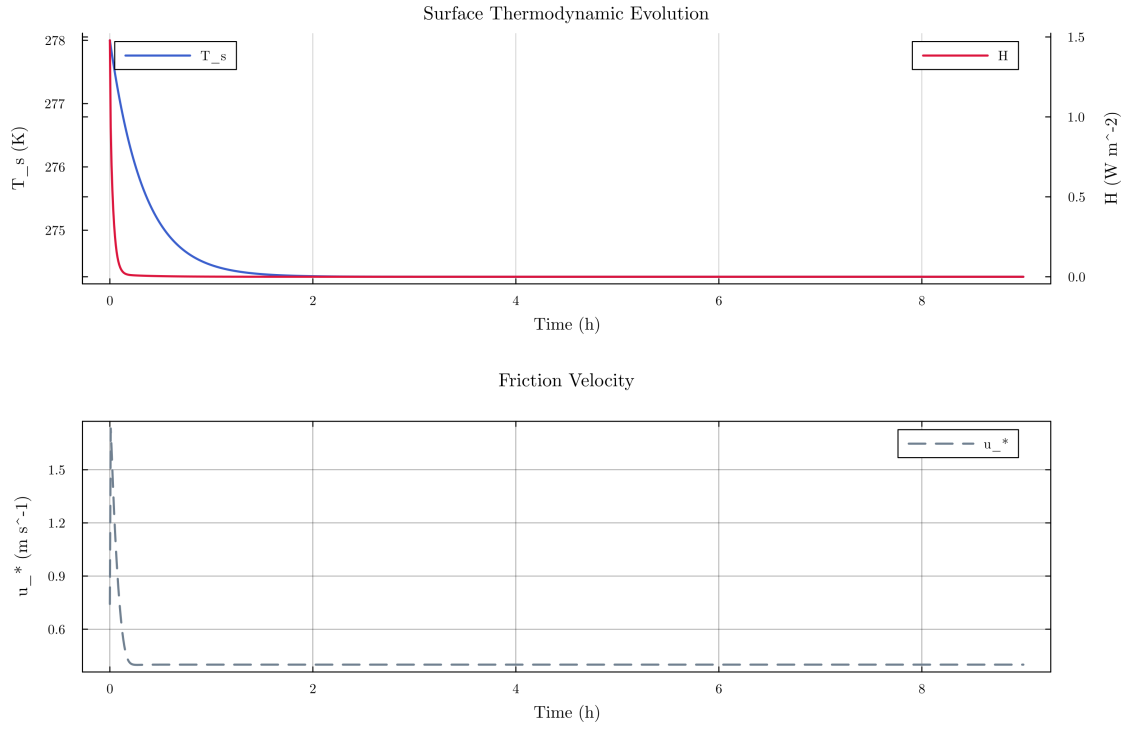


Figure 16: Time series of T_s , sensible heat flux, and friction velocity

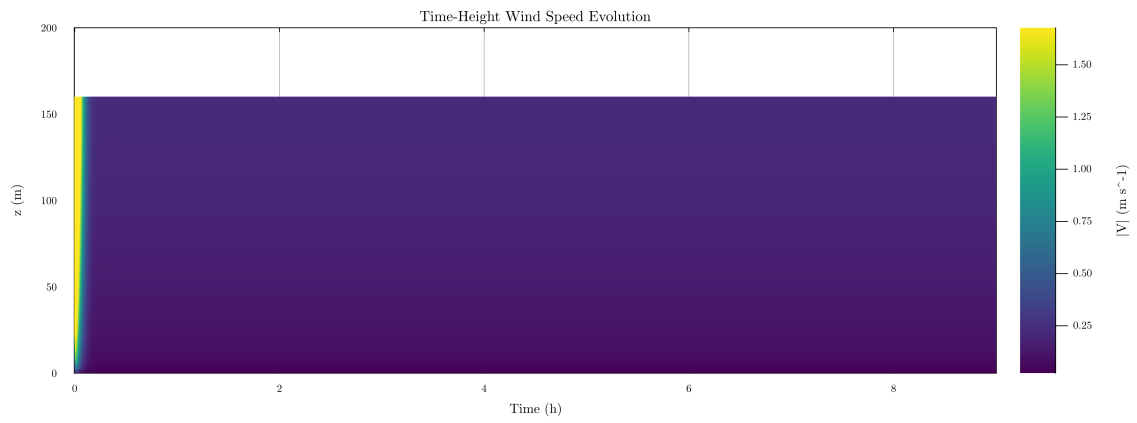


Figure 17: Time-height contour of wind speed with LLJ evolution

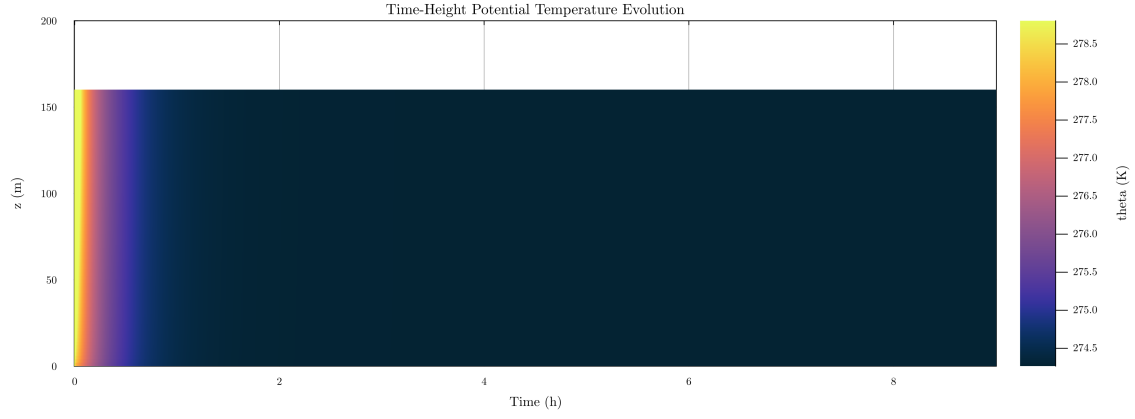


Figure 18: Time-height contour of potential temperature and inversion growth

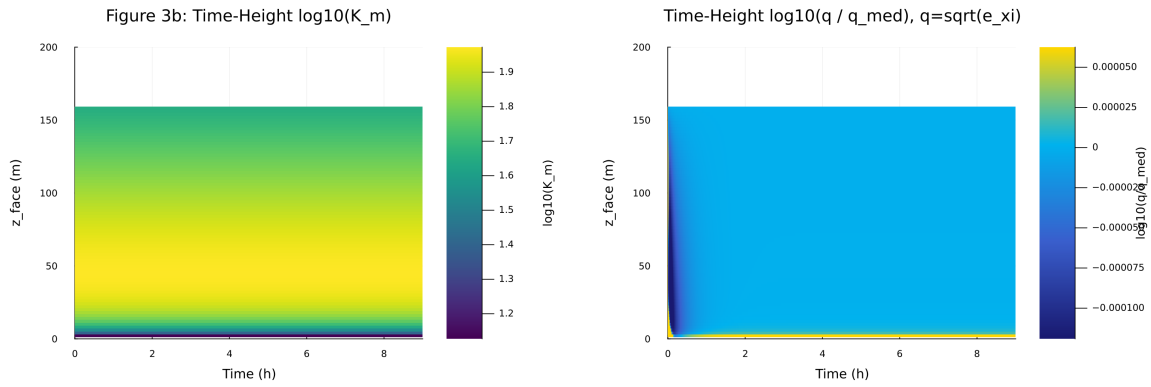


Figure 19: Vertical profiles of U , θ , and K_m at representative times

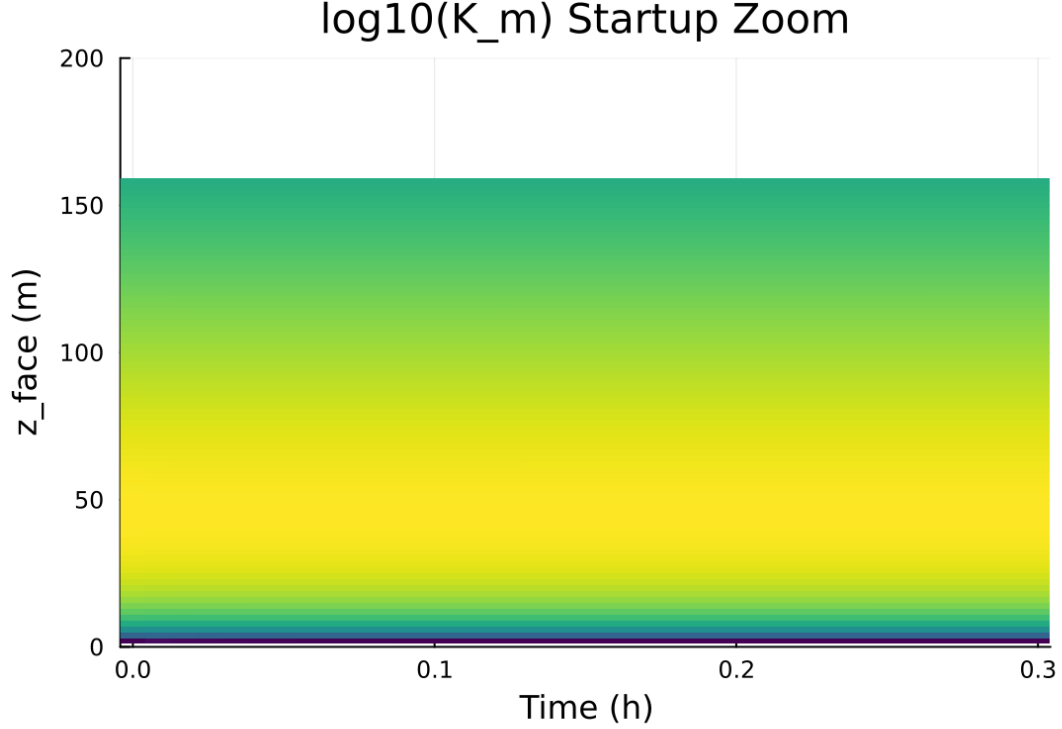


Figure 20: Surface energy budget components (R_n , H , G , storage)

6 Case 3: sheba 160x2m 12h report

7 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/SHEBA/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

7.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 100 / 0
- RHS evaluations: 49002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

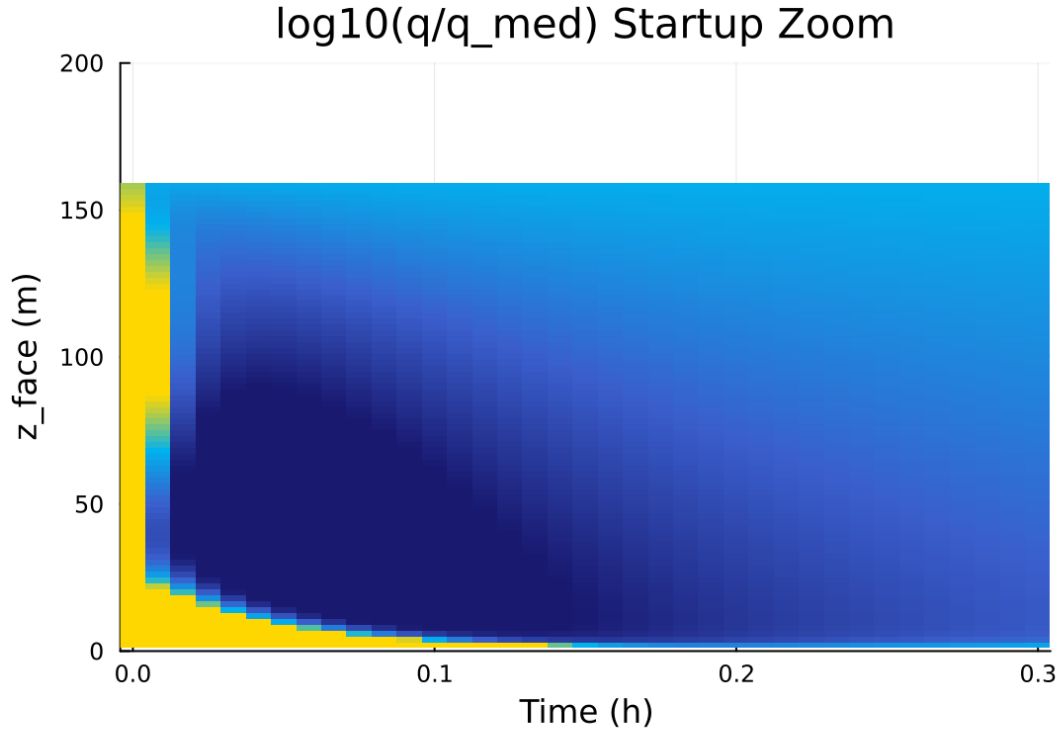


Figure 21: Phase portrait on regularized manifold (Delta vs e_{xi})

Table 5: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$7.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	$1.000\text{e-}04 \text{ m}$
Surface roughness (heat)	z_{0h}	$1.000\text{e-}05 \text{ m}$
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	257.000000 K
Deep soil temperature	T_{deep}	255.500000 K
Downward longwave forcing	$R_{\downarrow}$	$190.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	257.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

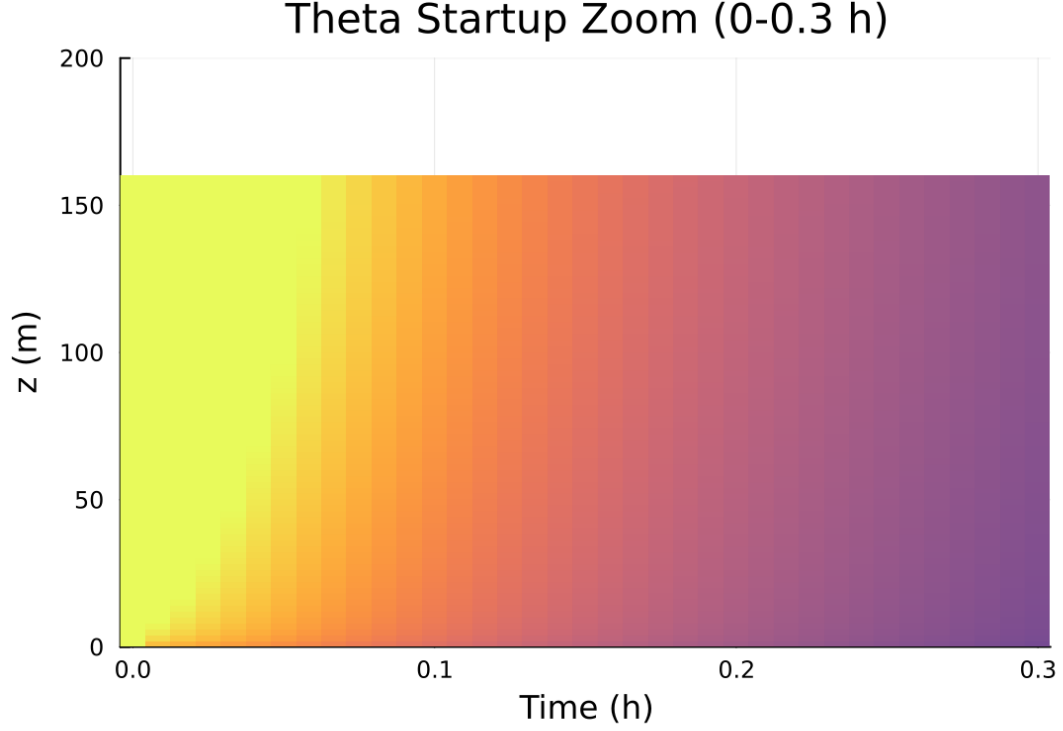


Figure 22: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)

7.2 Forcing and Boundary Conditions

7.3 GSPT Bifurcation & Manifold Calibration

Table 6: GSPT Normal Form Calibration Parameters for sheba

Metric	Symbol	Estimated Value
Report / Calibration Status	—	invalid_fit
Estimated Fold Richardson Number	$\hat{Ri}_{\text{fold}}$	NA
95% MBB Confidence Interval	CI_{95}	[NA, NA]
Transcritical Ignition Proxy	$\hat{Ri}_{\text{critical}}$	0.000000
Hysteresis Width	ΔRi_H	NA
Scaling Exponent	$\hat{\gamma}$	NA
Normal Form Amplitude	$\hat{c}$	NA

7.4 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [13.028341, 141.433765] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-4.828 \times 10^{-6}, 3569.528439]$

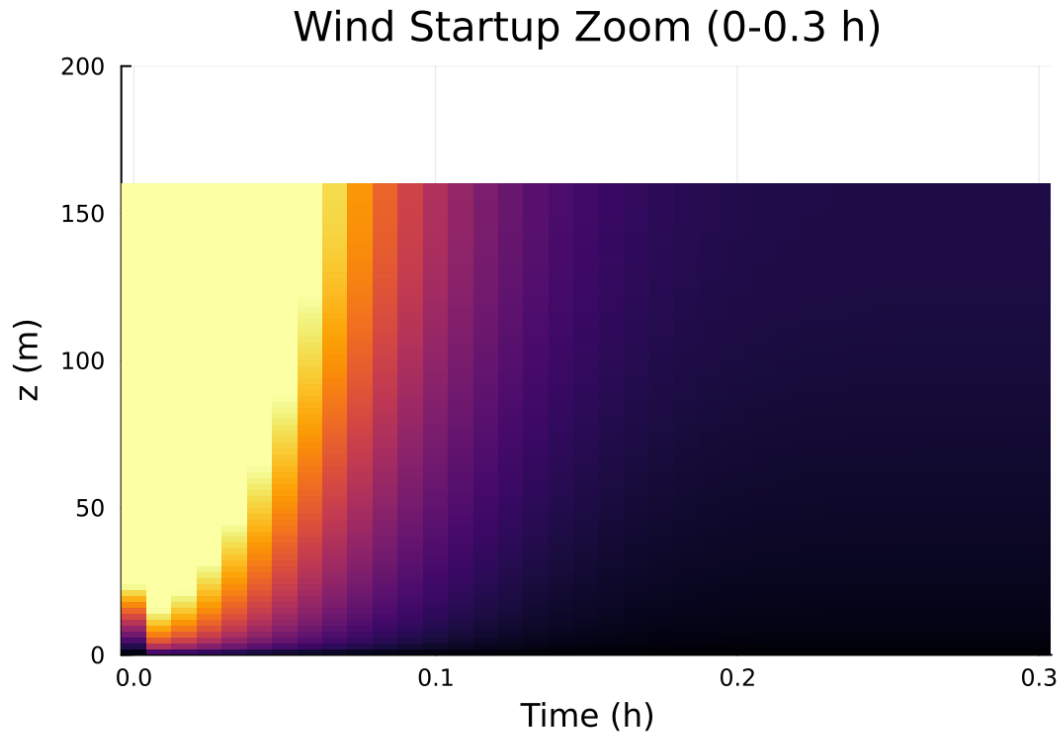


Figure 23: Fold proximity diagnostics versus time

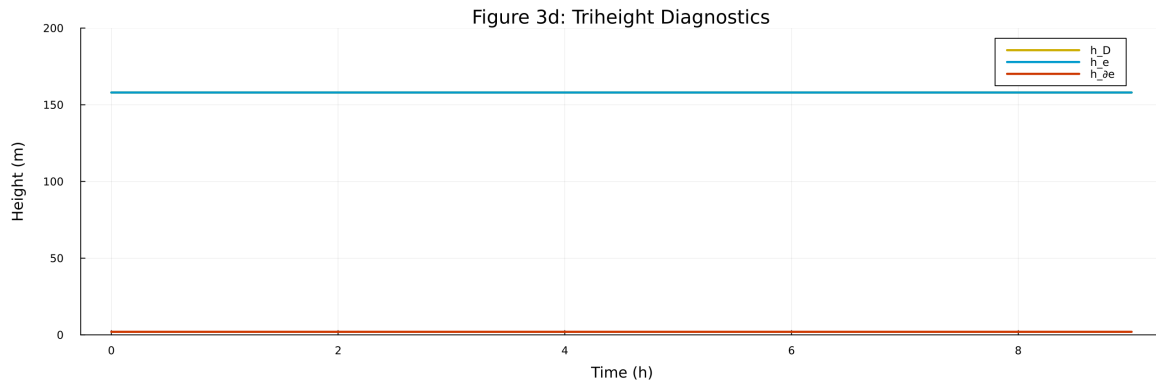


Figure 24: fig03d triheight timeseries.png

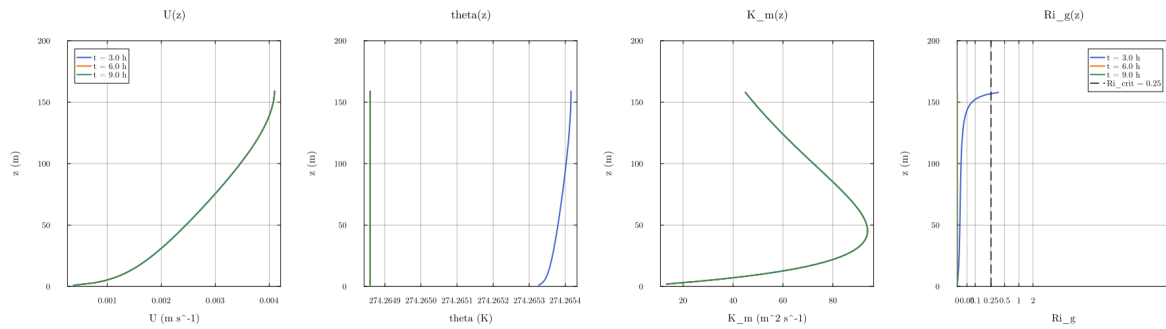


Figure 25: fig04 profiles u theta km.png

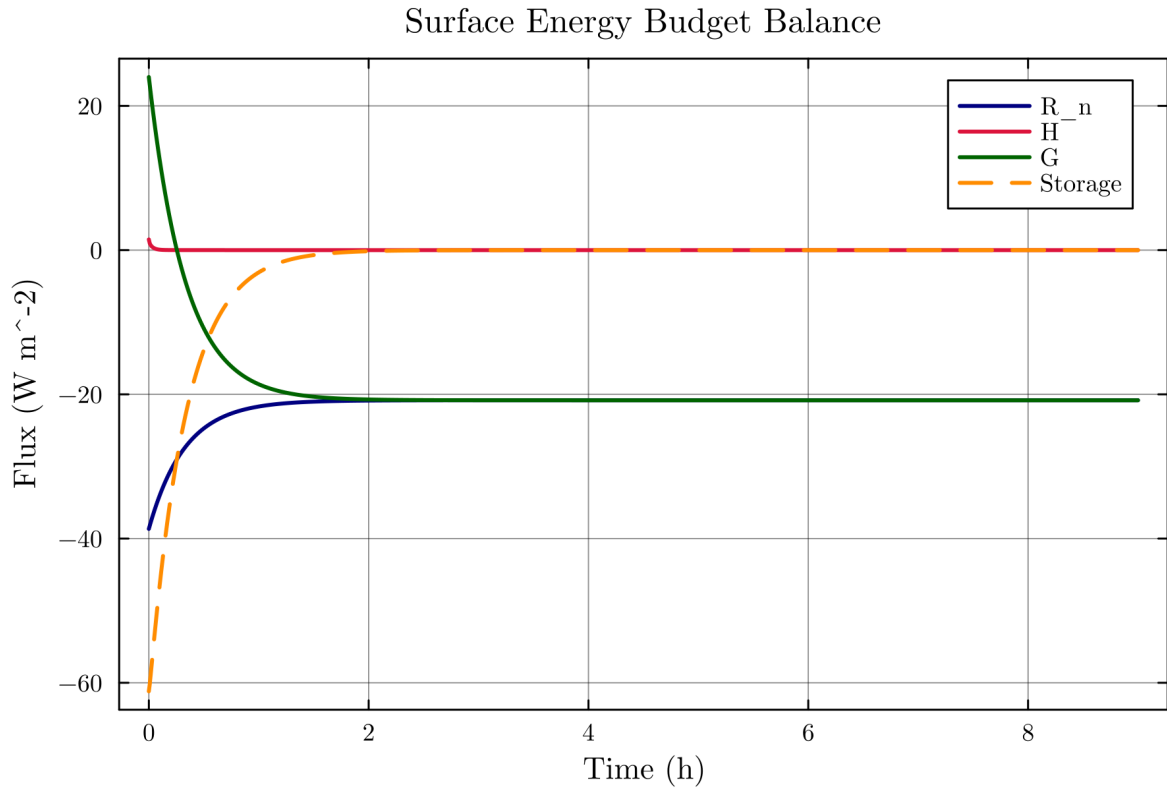


Figure 26: fig05 surface energy budget.png

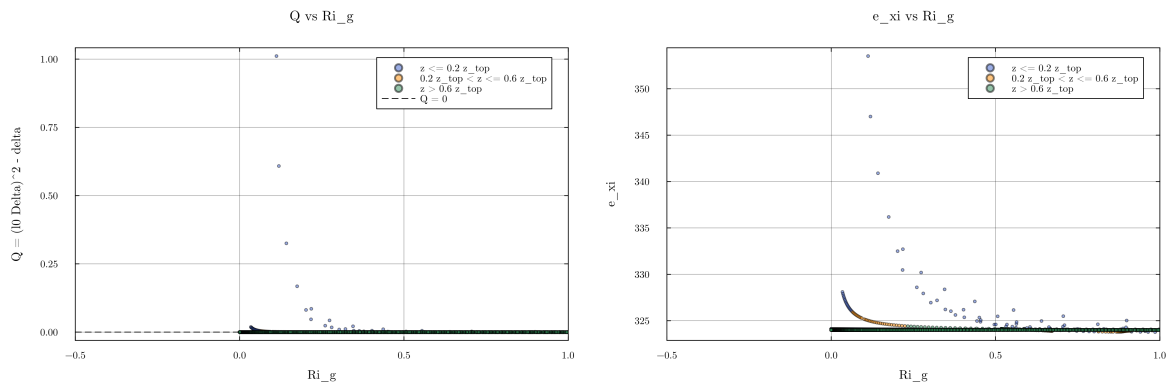


Figure 27: fig06 phase delta exi.png

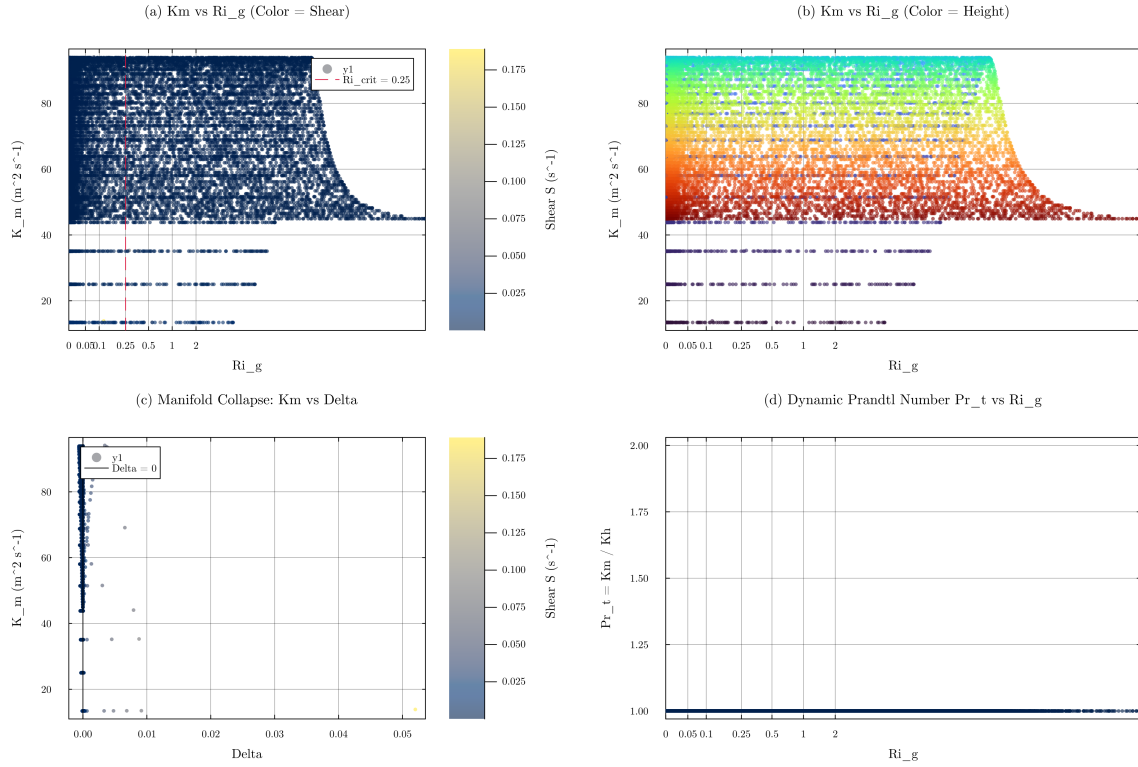


Figure 28: fig07 closure manifold.png

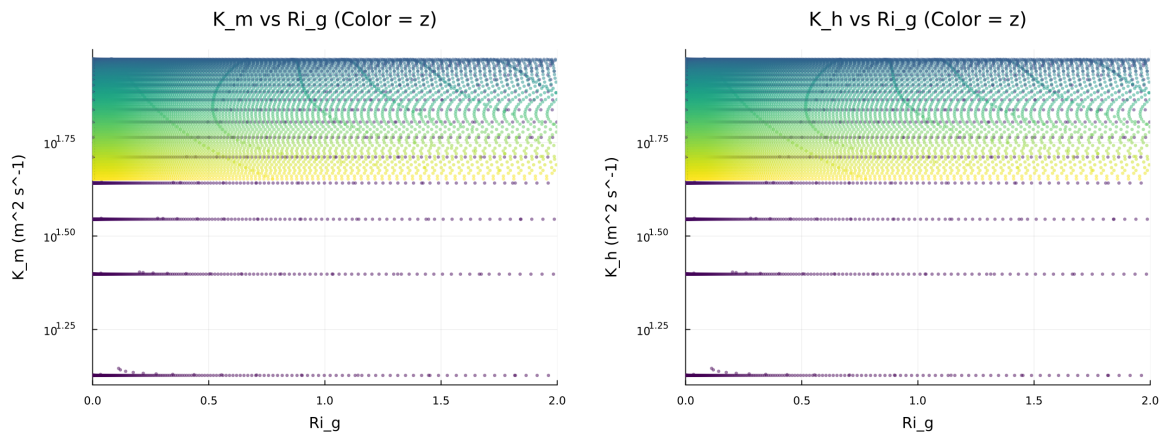


Figure 29: fig07 diffusivity vs ri.png

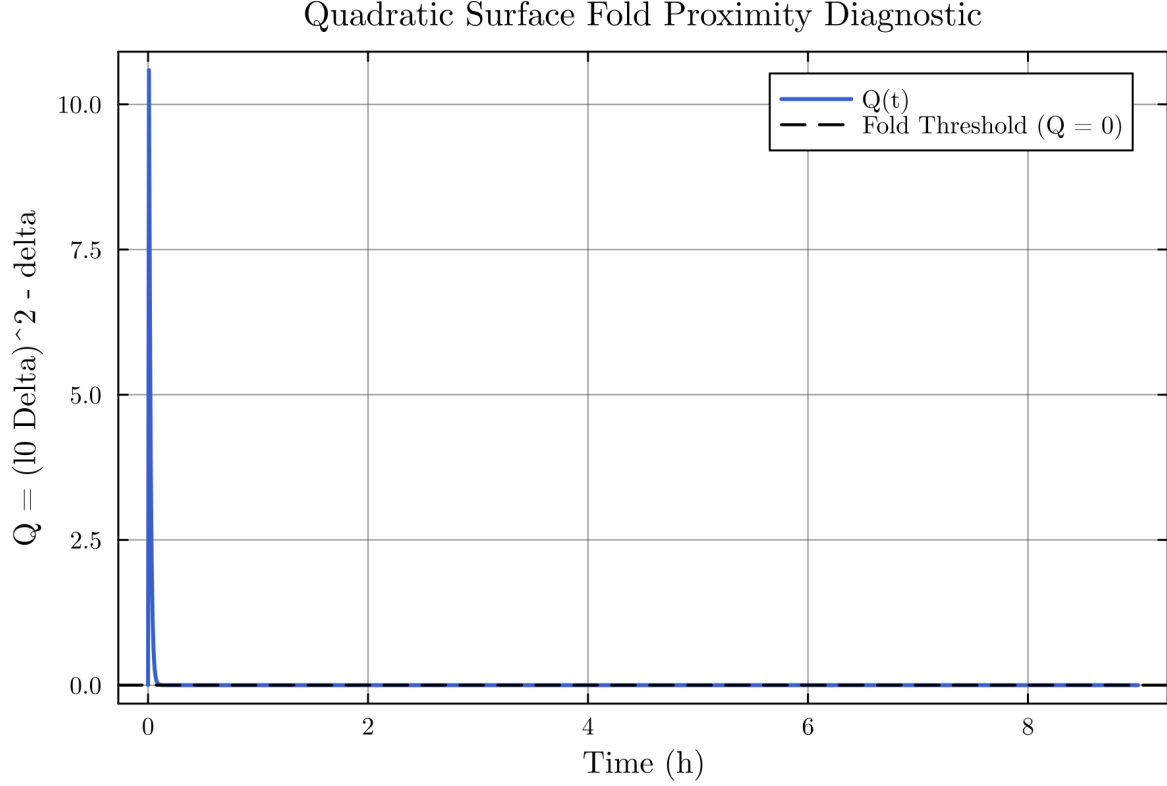


Figure 30: fig08 fold proximity.png

7.5 Generated Artifacts

- Time-series CSV: `results/sheba/time_series.csv`
- Diagnostic summary JSON: `results/SHEBA/summary.json`
- Payload JLD2: `results/sheba/payload.jld2`

7.6 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Closure manifold diagnostics (K_m vs Ri_g and K_m vs Δ)
8. Fold proximity diagnostics versus time

7.7 Included Plots

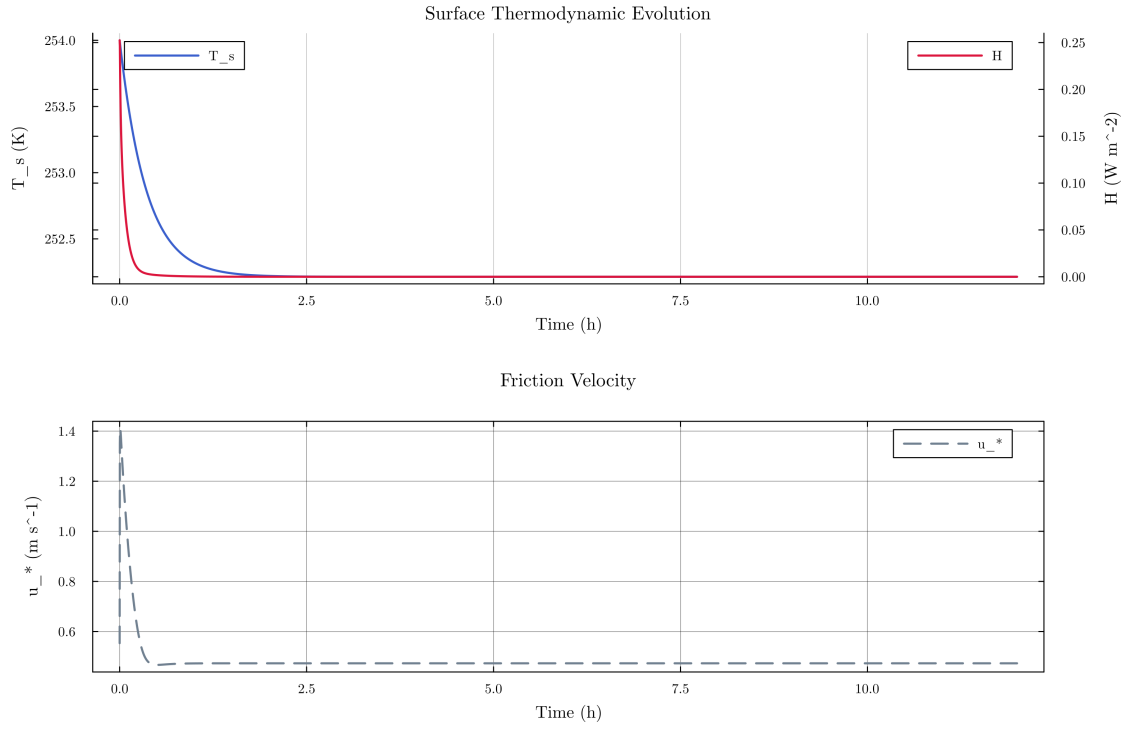


Figure 31: Time series of T_s , sensible heat flux, and friction velocity

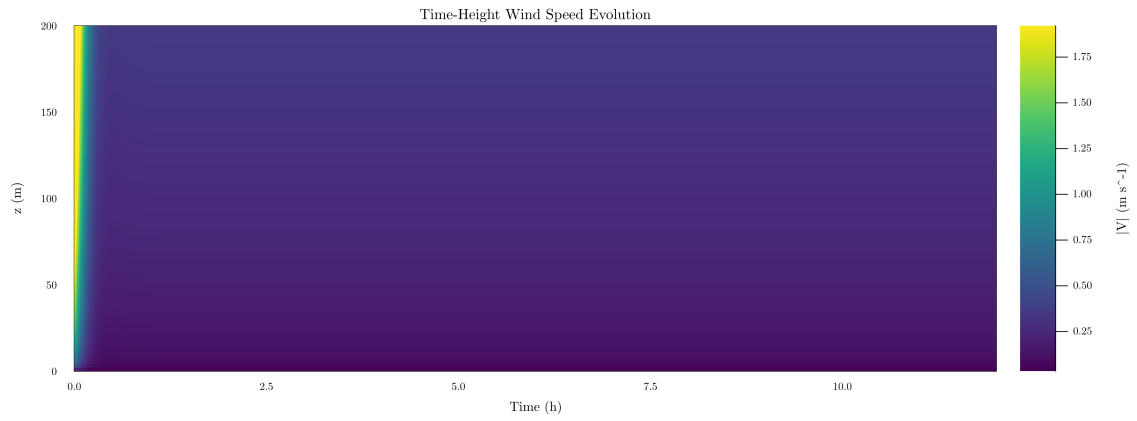


Figure 32: Time-height contour of wind speed with LLJ evolution

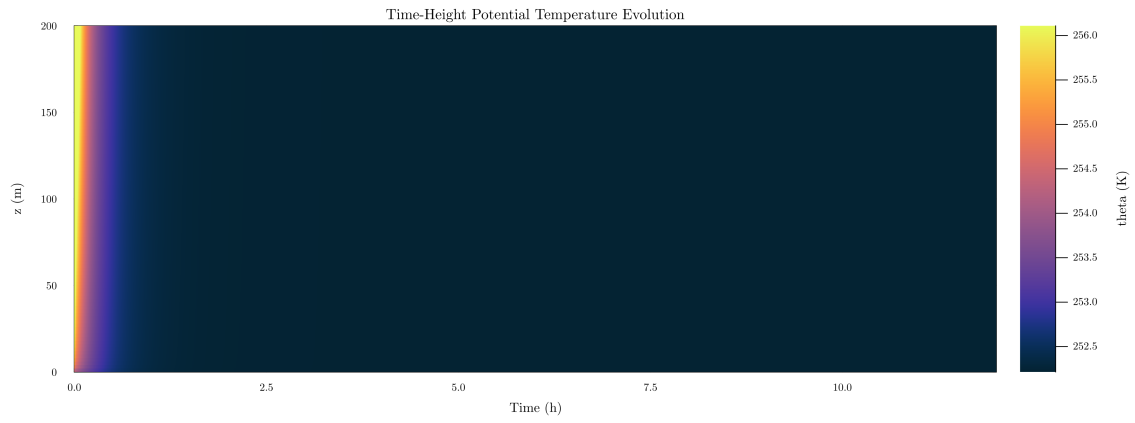


Figure 33: Time-height contour of potential temperature and inversion growth

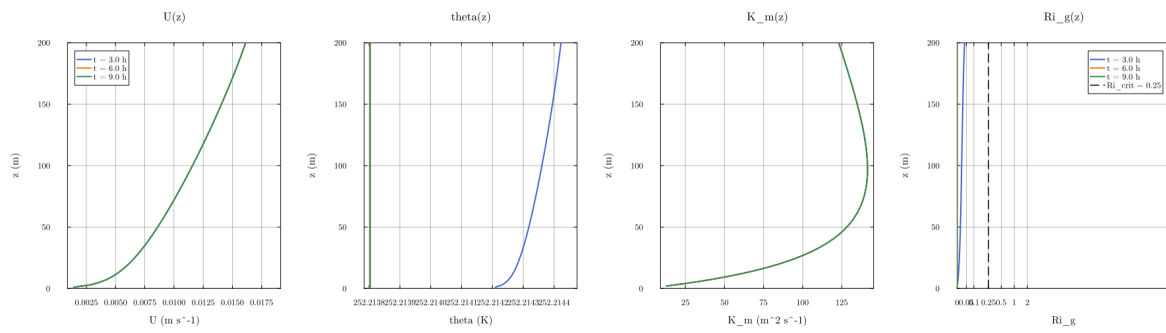


Figure 34: Vertical profiles of U , θ , and K_m at representative times

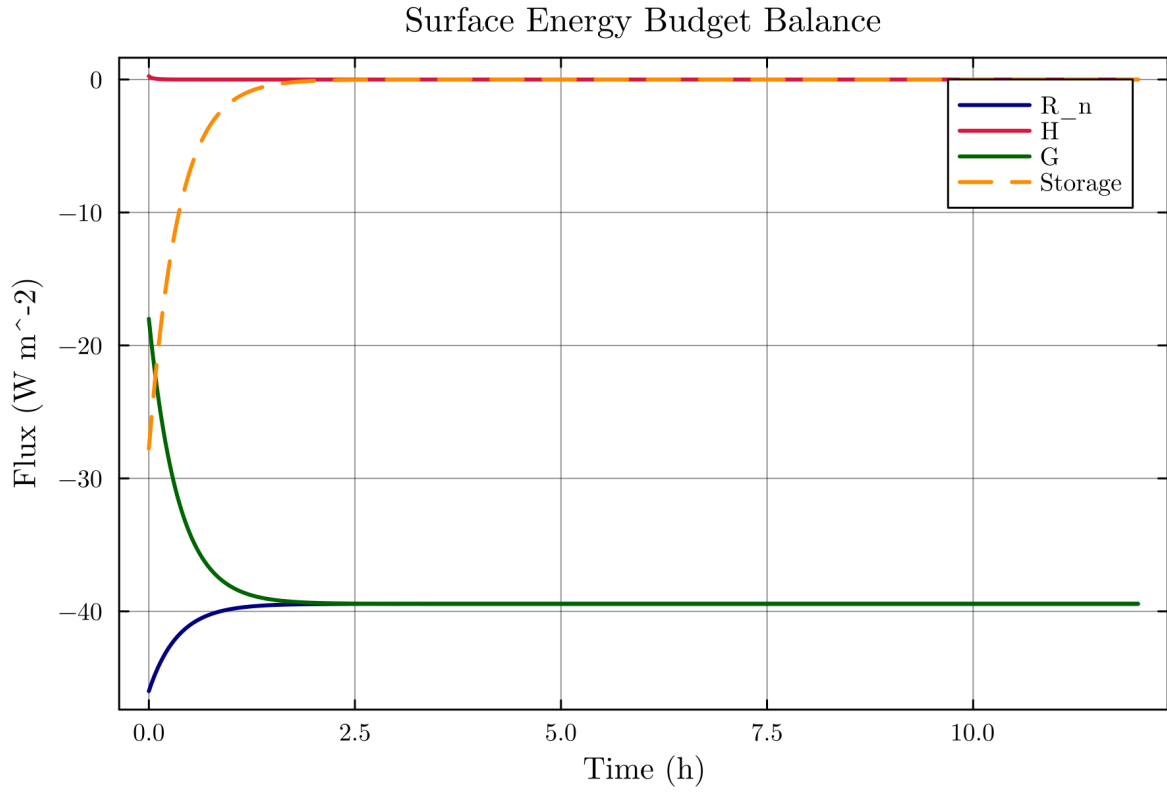


Figure 35: Surface energy budget components (R_n , H , G , storage)

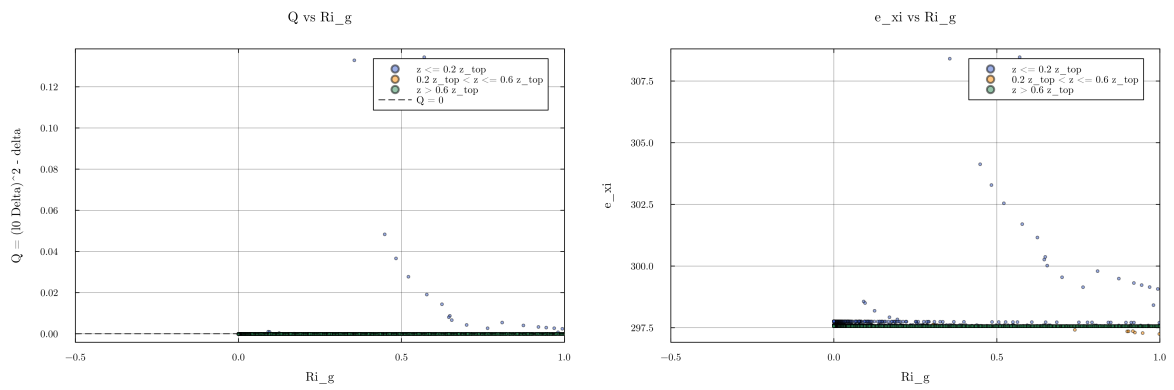


Figure 36: Phase portrait on regularized manifold (Δ vs e_{xi})

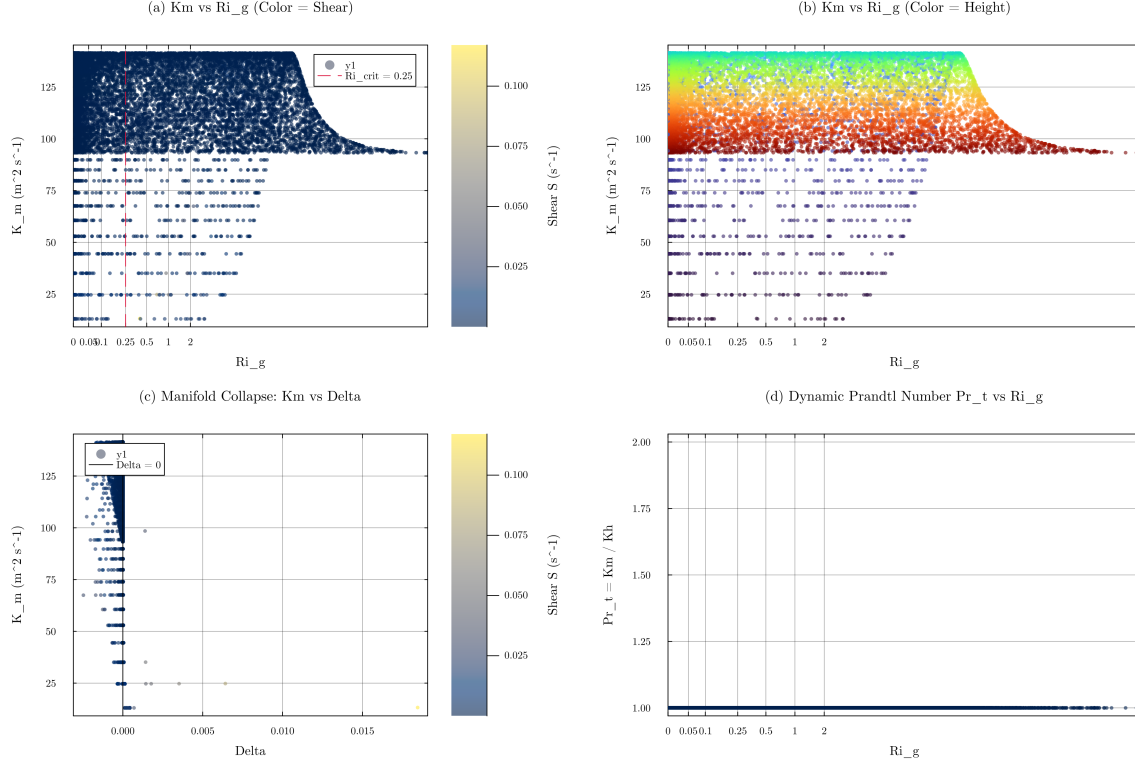


Figure 37: Closure manifold diagnostics (K_m vs Ri_g and K_m vs Δ)

Quadratic Surface Fold Proximity Diagnostic

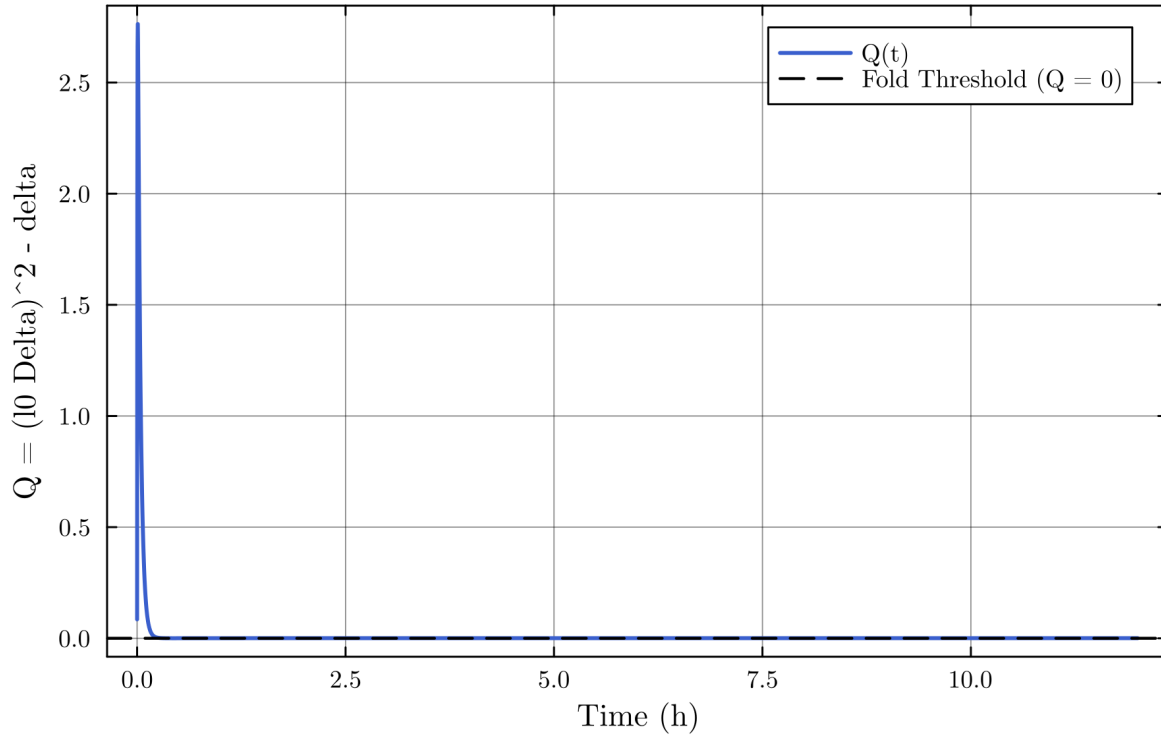


Figure 38: Fold proximity diagnostics versus time

8 Case 4: gabls1 200x2m 12h report

9 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

9.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 102 / 0
- RHS evaluations: 25502
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

9.2 Forcing and Boundary Conditions

Table 7: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.100000 m
Surface roughness (heat)	z_{0h}	0.010000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\min, \text{surf}}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

9.3 GSPT Bifurcation & Manifold Calibration

9.4 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 94.819611%
- Diffusivity bounds: $K_m \in [11.165528, 78.288199] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-2.750 \times 10^{-7}, 1888.099836]$

Table 8: GSPT Normal Form Calibration Parameters for gabls1

Metric	Symbol	Estimated Value
Report / Calibration Status	—	invalid_fit
Estimated Fold Richardson Number	$\hat{Ri}_{\text{fold}}$	NA
95% MBB Confidence Interval	CI ₉₅	[NA, NA]
Transcritical Ignition Proxy	$\hat{Ri}_{\text{critical}}$	0.000000
Hysteresis Width	ΔRi_H	NA
Scaling Exponent	$\hat{\gamma}$	NA
Normal Form Amplitude	$\hat{c}$	NA

9.5 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

9.6 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)
8. Fold proximity diagnostics versus time

9.7 Included Plots

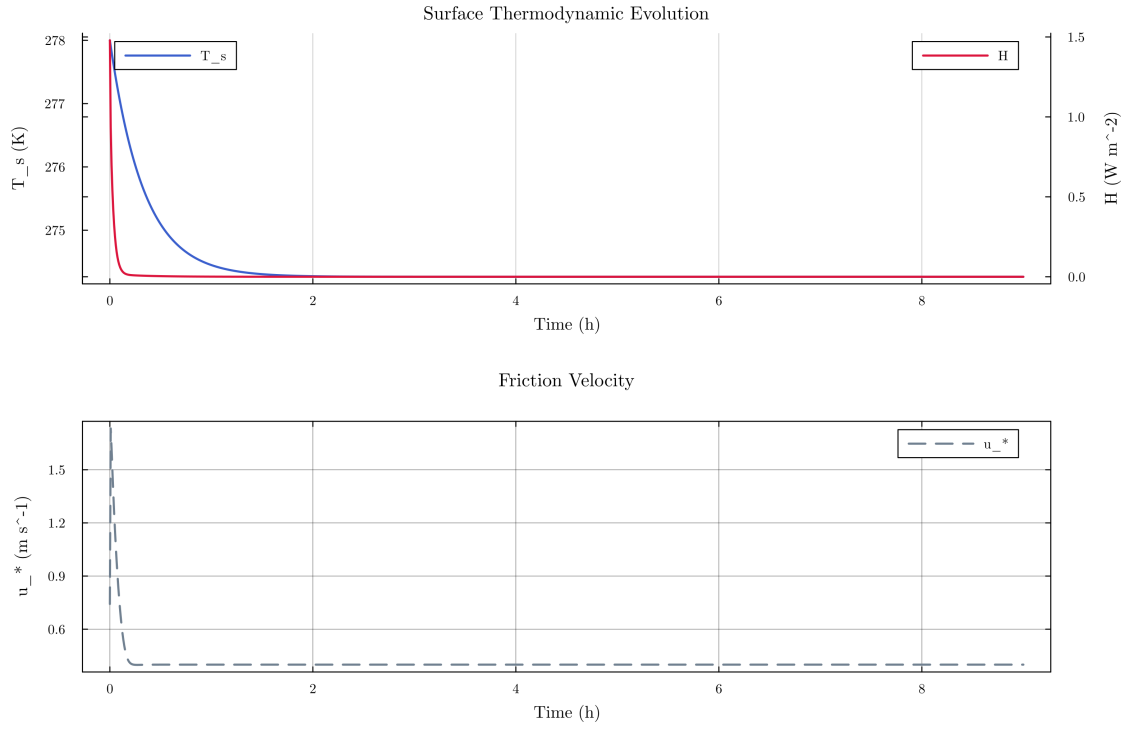


Figure 39: Time series of T_s , sensible heat flux, and friction velocity

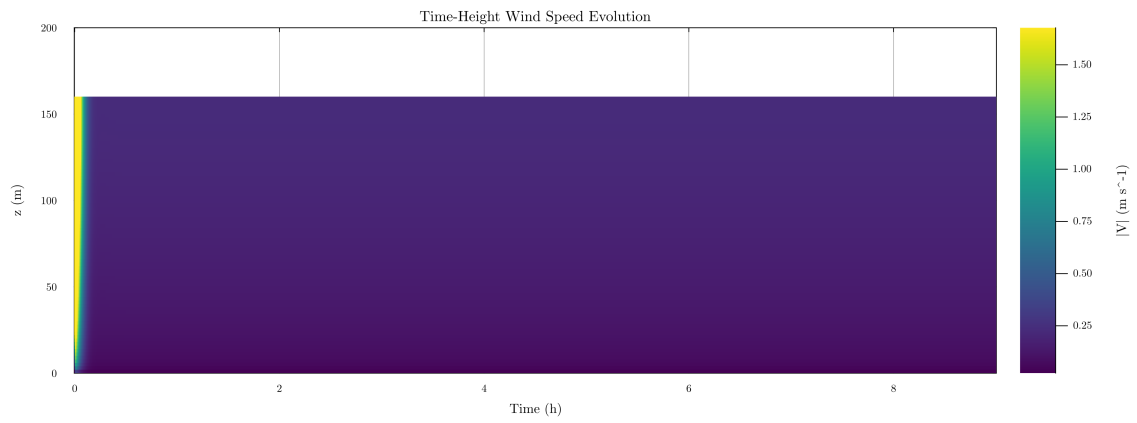


Figure 40: Time-height contour of wind speed with LLJ evolution

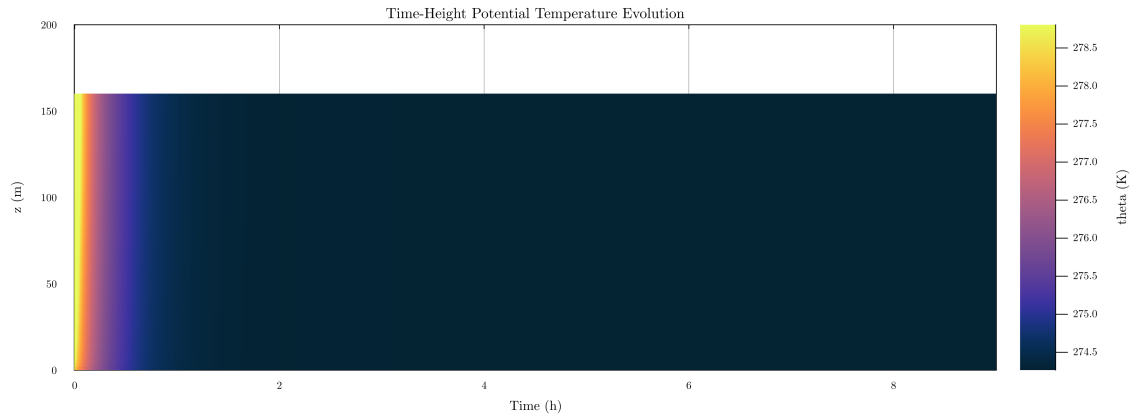


Figure 41: Time-height contour of potential temperature and inversion growth

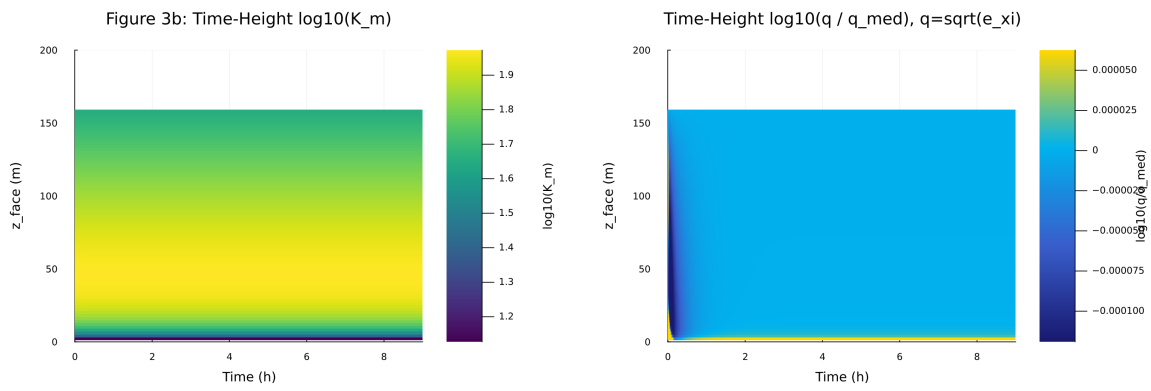


Figure 42: Vertical profiles of U , θ , and K_m at representative times

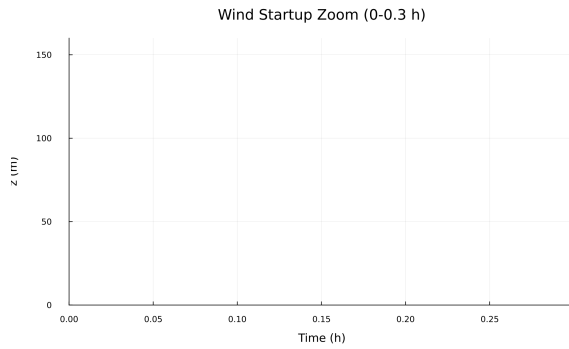


Figure 43: Surface energy budget components (R_n , H , G , storage)

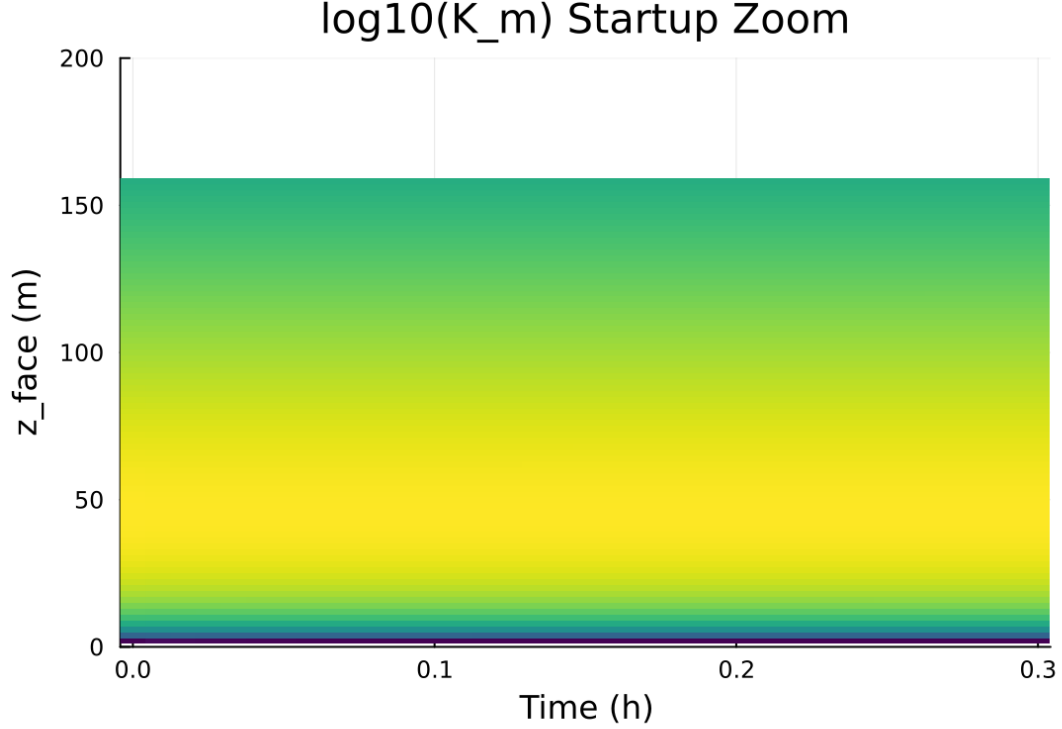


Figure 44: Phase portrait on regularized manifold (Delta vs e_xi)

10 Case 5: gabls1 80x2m 9h report

11 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

11.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 102 / 0
- RHS evaluations: 25502
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

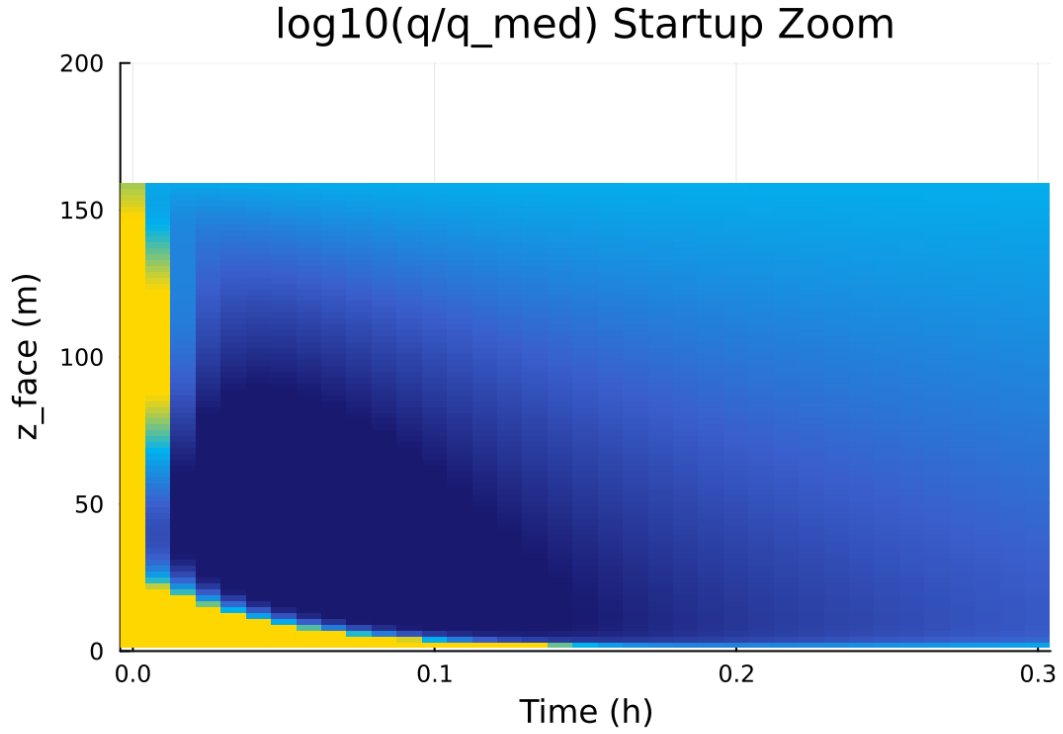


Figure 45: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.100000 m
Surface roughness (heat)	z_{0h}	0.010000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

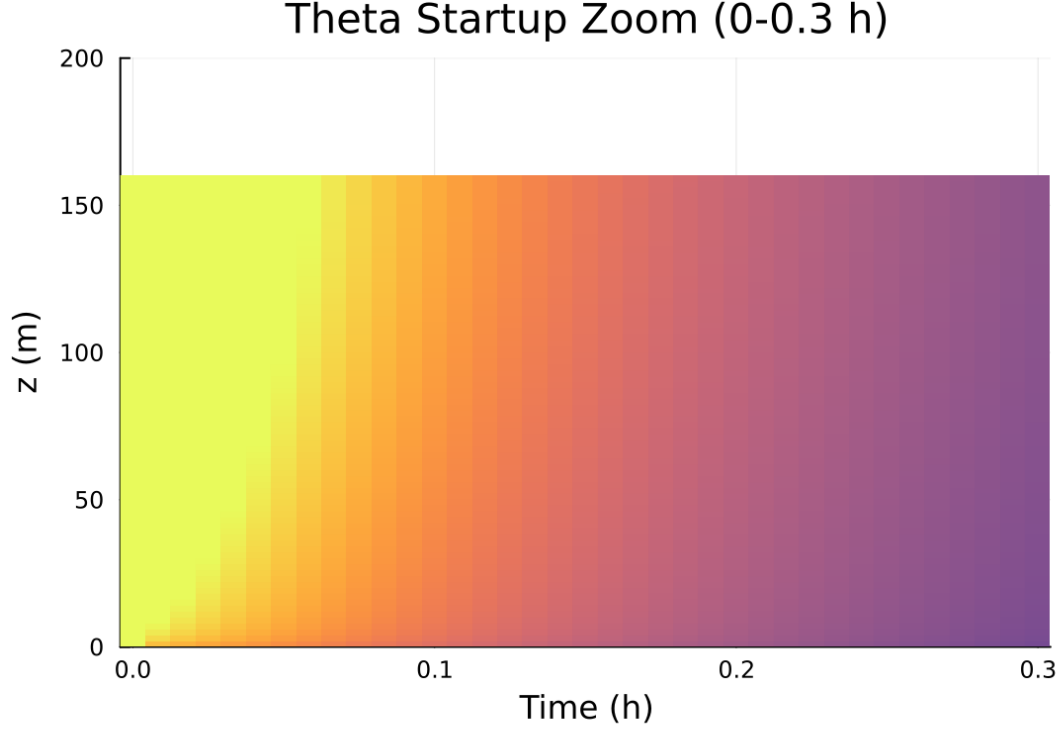


Figure 46: Fold proximity diagnostics versus time

11.2 Forcing and Boundary Conditions

11.3 GSPT Bifurcation & Manifold Calibration

Table 10: GSPT Normal Form Calibration Parameters for gabls1

Metric	Symbol	Estimated Value
Report / Calibration Status	—	invalid_fit
Estimated Fold Richardson Number	$\hat{Ri}_{\text{fold}}$	NA
95% MBB Confidence Interval	CI_{95}	[NA, NA]
Transcritical Ignition Proxy	$\hat{Ri}_{\text{critical}}$	0.000000
Hysteresis Width	ΔRi_H	NA
Scaling Exponent	$\hat{\gamma}$	NA
Normal Form Amplitude	$\hat{c}$	NA

11.4 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 94.819611%
- Diffusivity bounds: $K_m \in [11.165528, 78.288199] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-2.750 \times 10^{-7}, 1888.099836]$

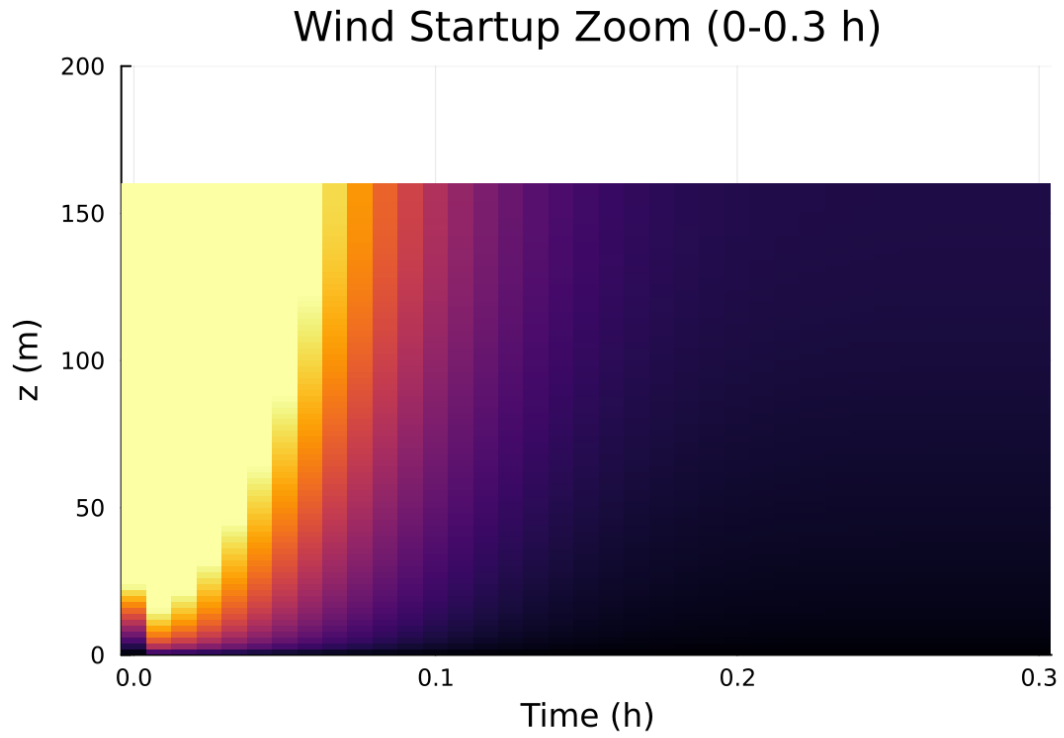


Figure 47: fig03c wind startup zoom.png

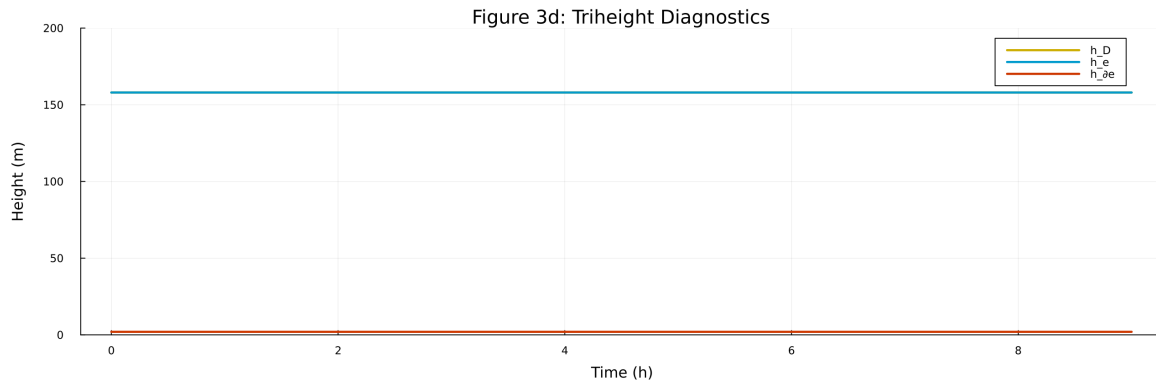


Figure 48: fig03d triheight timeseries.png

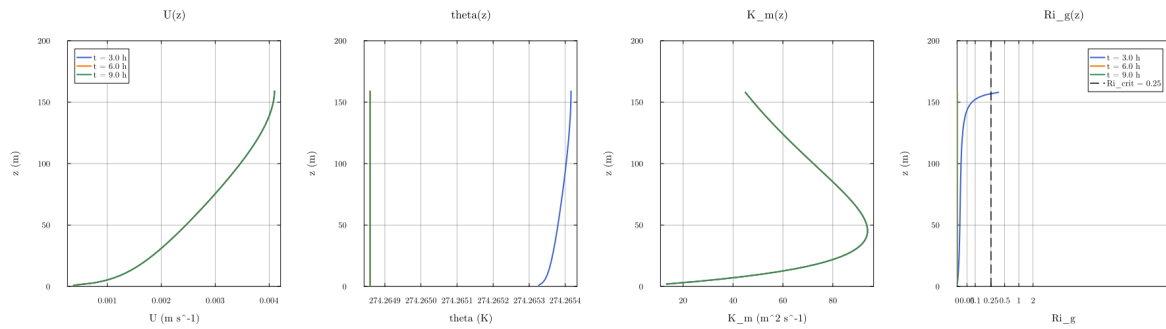


Figure 49: fig04 profiles u theta km.png

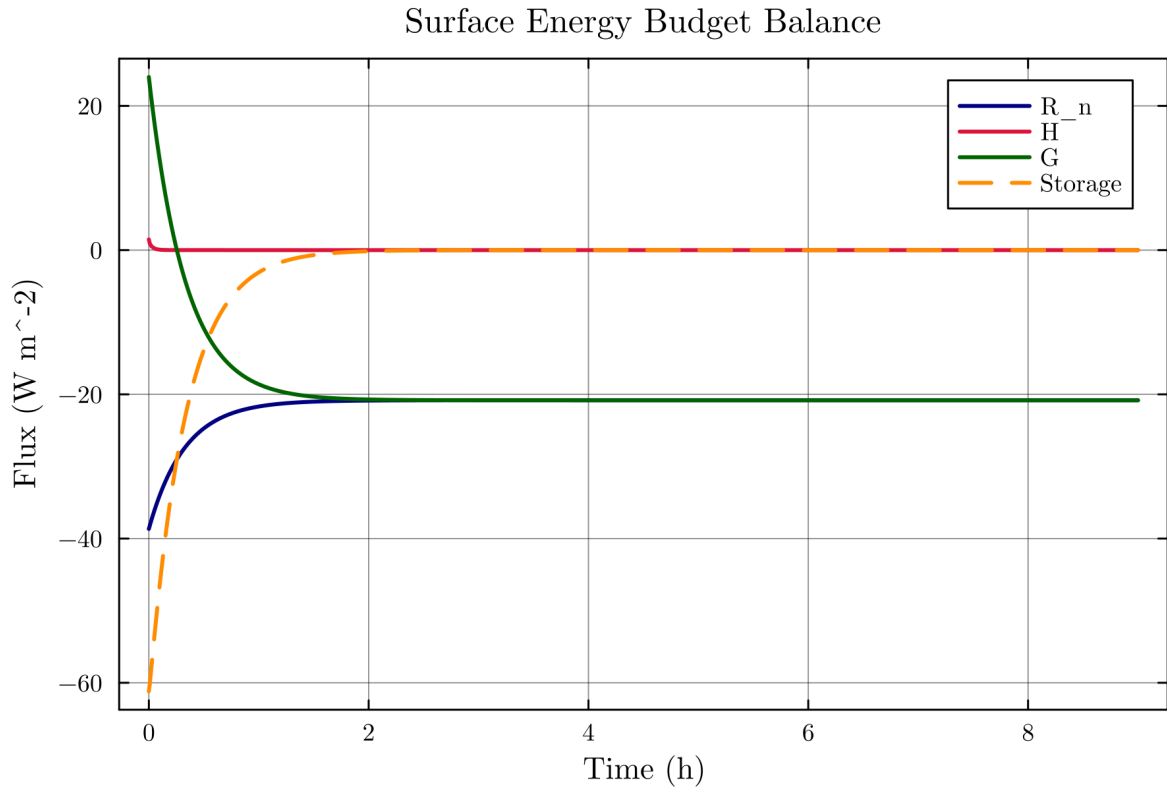


Figure 50: fig05 surface energy budget.png

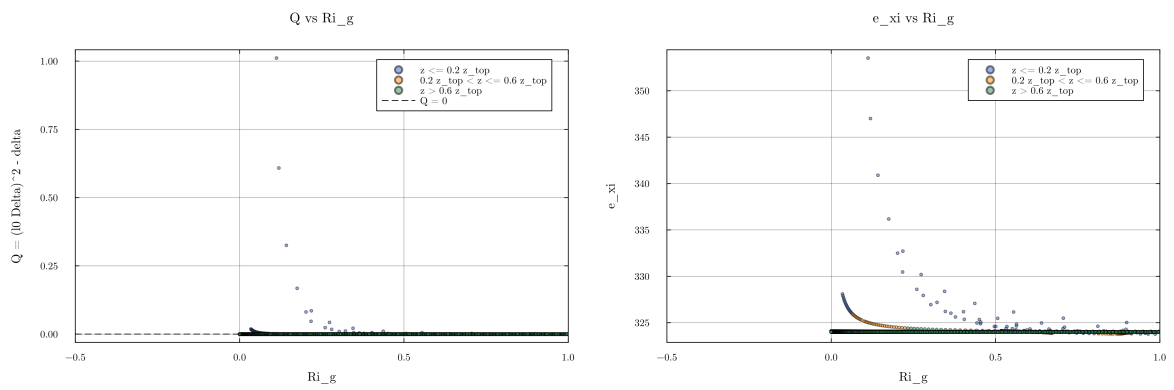


Figure 51: fig06 phase delta exi.png

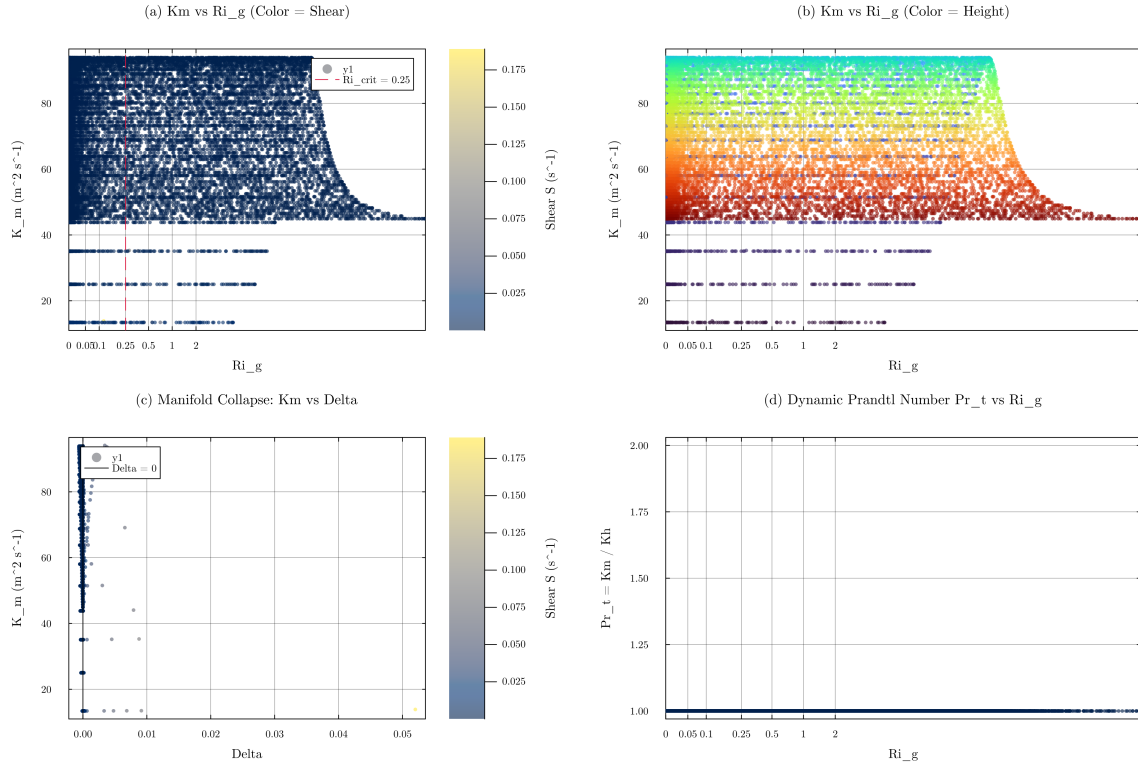


Figure 52: fig07 closure manifold.png

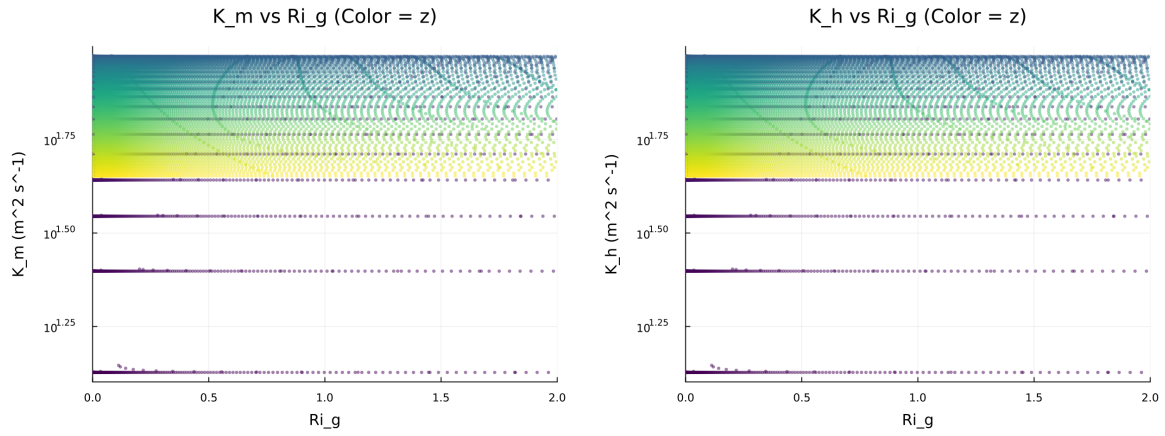


Figure 53: fig07 diffusivity vs ri.png

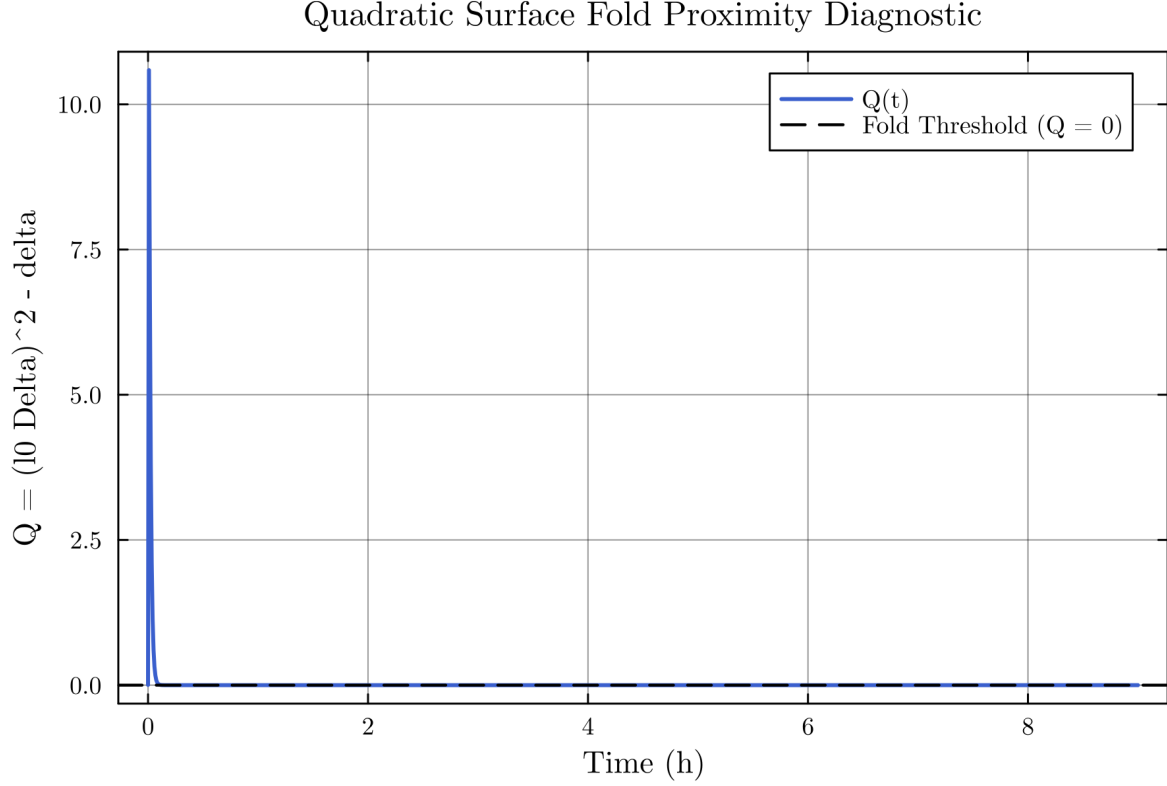


Figure 54: fig08 fold proximity.png

11.5 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

11.6 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Closure manifold diagnostics (K_m vs Ri_g and K_m vs Δ)
8. Fold proximity diagnostics versus time

11.7 Included Plots

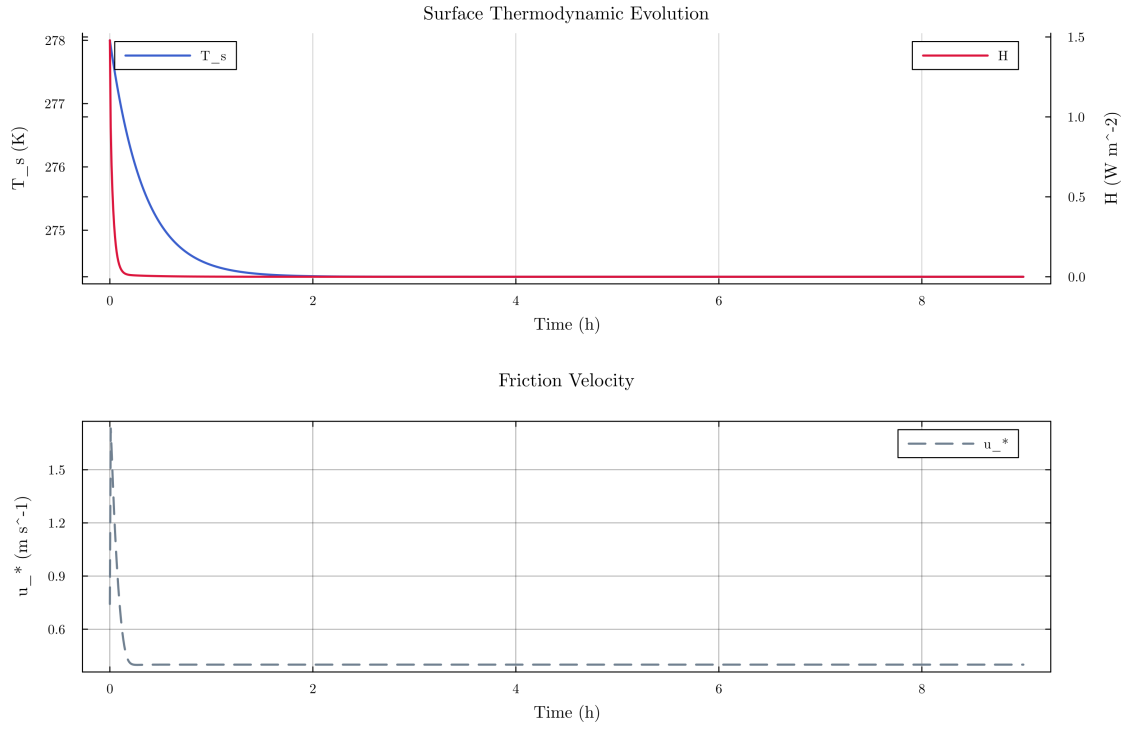


Figure 55: Time series of T_s , sensible heat flux, and friction velocity

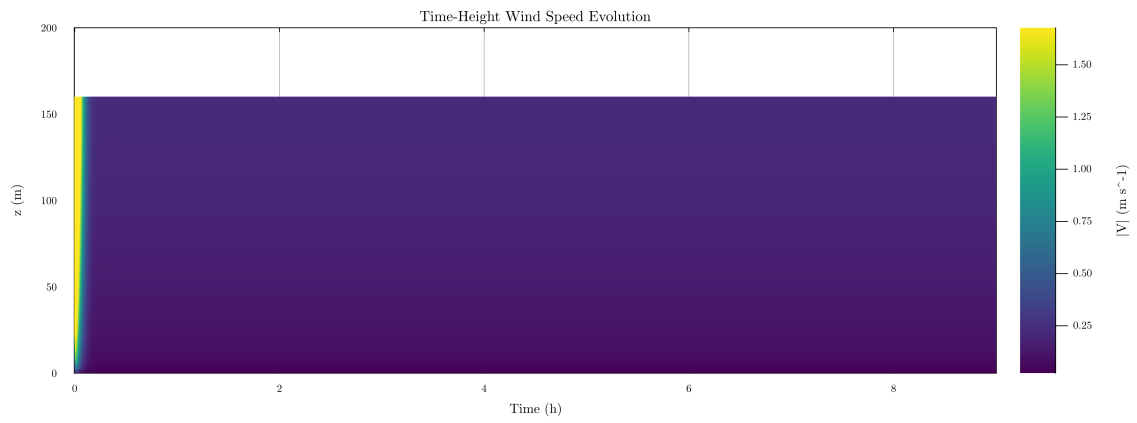


Figure 56: Time-height contour of wind speed with LLJ evolution

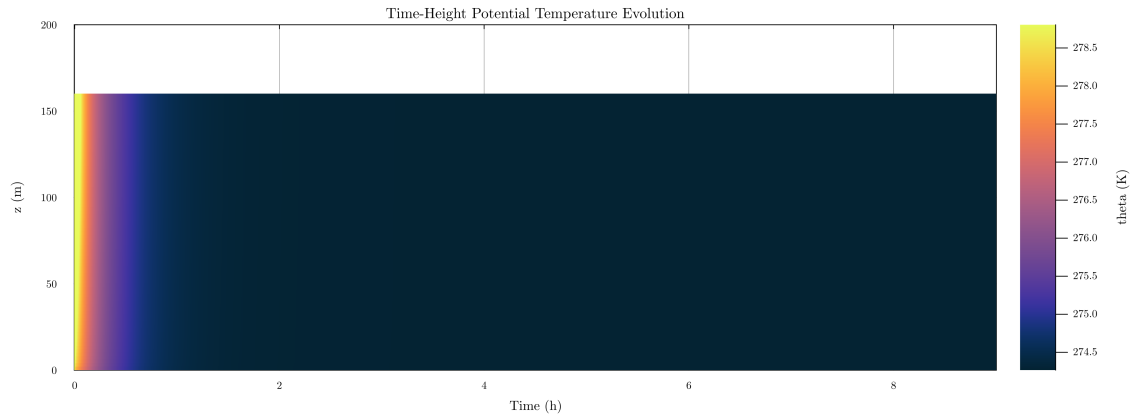


Figure 57: Time-height contour of potential temperature and inversion growth

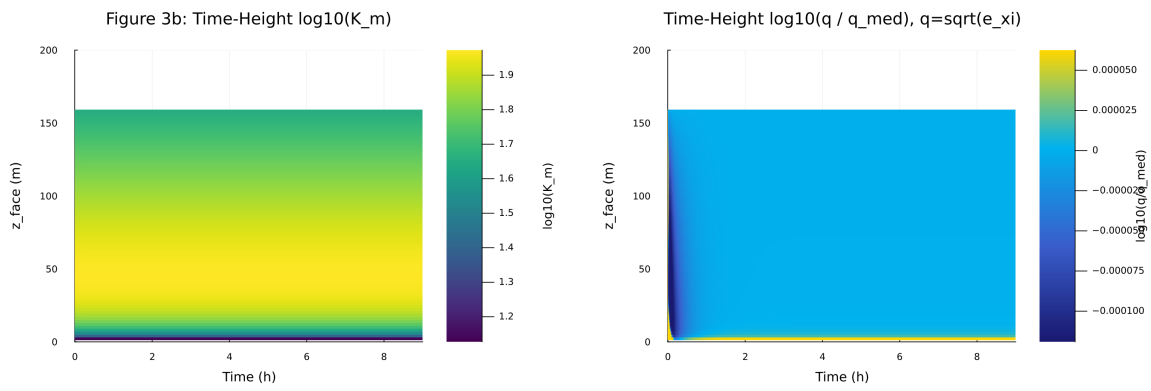


Figure 58: Vertical profiles of U, theta, and Km at representative times

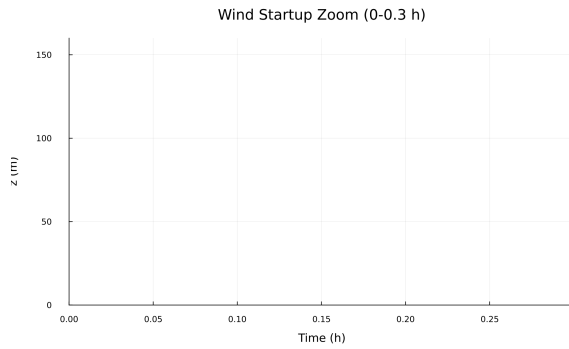


Figure 59: Surface energy budget components (R_n , H , G , storage)

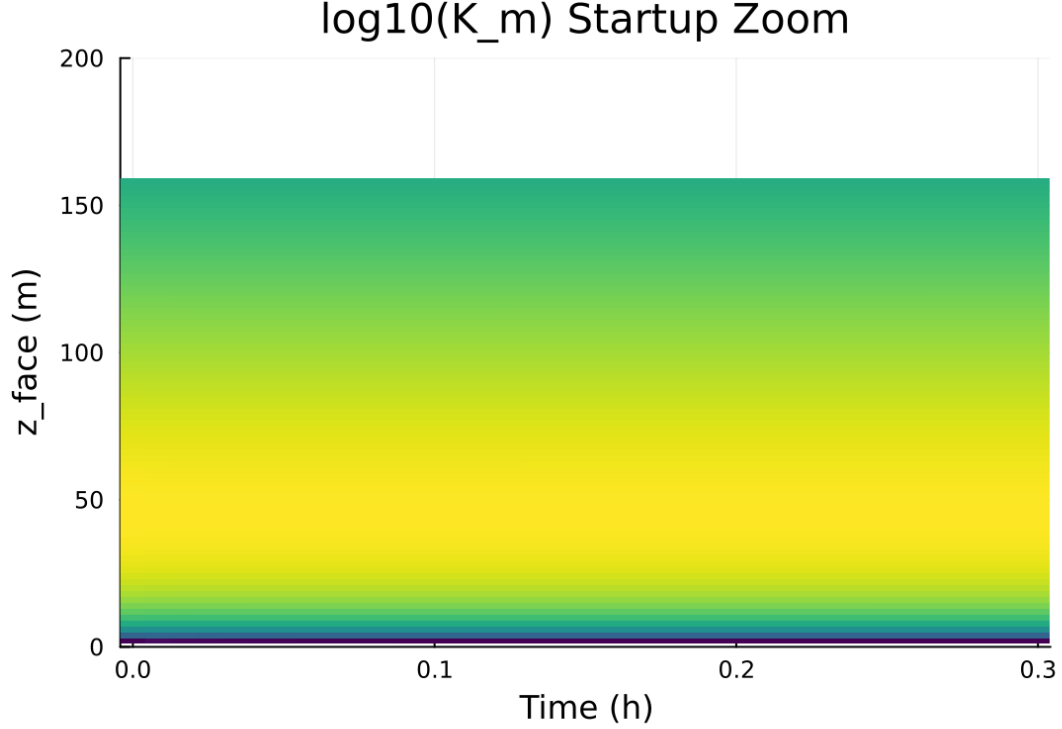


Figure 60: Phase portrait on regularized manifold (Delta vs e_{xi})

12 Case 6: idealized sbl 80x2m 9h report

13 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

13.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

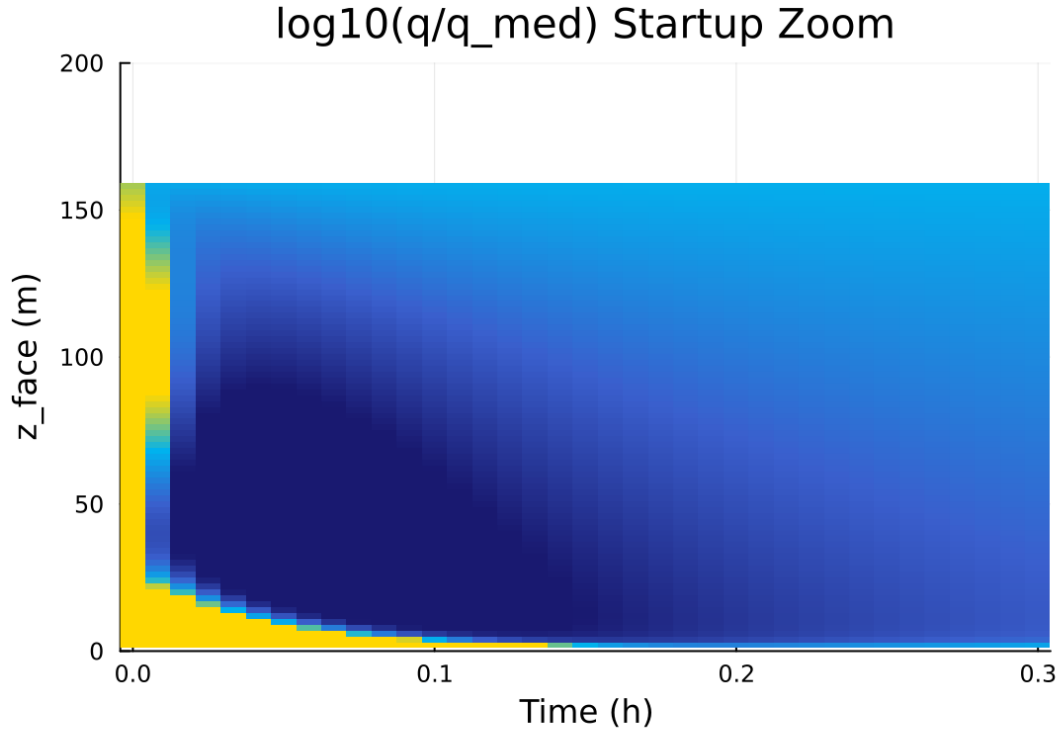


Figure 61: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)

Table 11: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

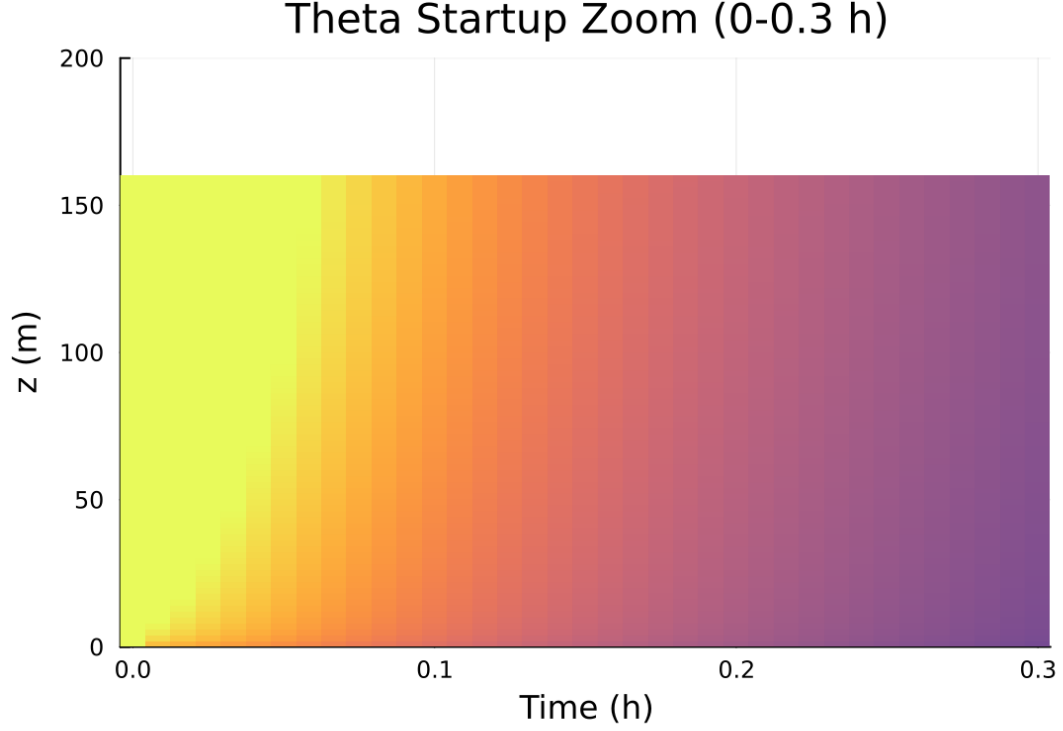


Figure 62: Fold proximity diagnostics versus time

13.2 Forcing and Boundary Conditions

13.3 GSPT Bifurcation & Manifold Calibration

Table 12: GSPT Normal Form Calibration Parameters for idealized_sbl

Metric	Symbol	Estimated Value
Report / Calibration Status	—	not_available
Estimated Fold Richardson Number	$\hat{Ri}_{\text{fold}}$	NA
95% MBB Confidence Interval	CI_{95}	[NA, NA]
Transcritical Ignition Proxy	$\hat{Ri}_{\text{critical}}$	NA
Hysteresis Width	ΔRi_H	NA
Scaling Exponent	$\hat{\gamma}$	NA
Normal Form Amplitude	$\hat{c}$	NA

13.4 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399988, 94.108850] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-2.012 \times 10^{-6}, 1552.767352]$

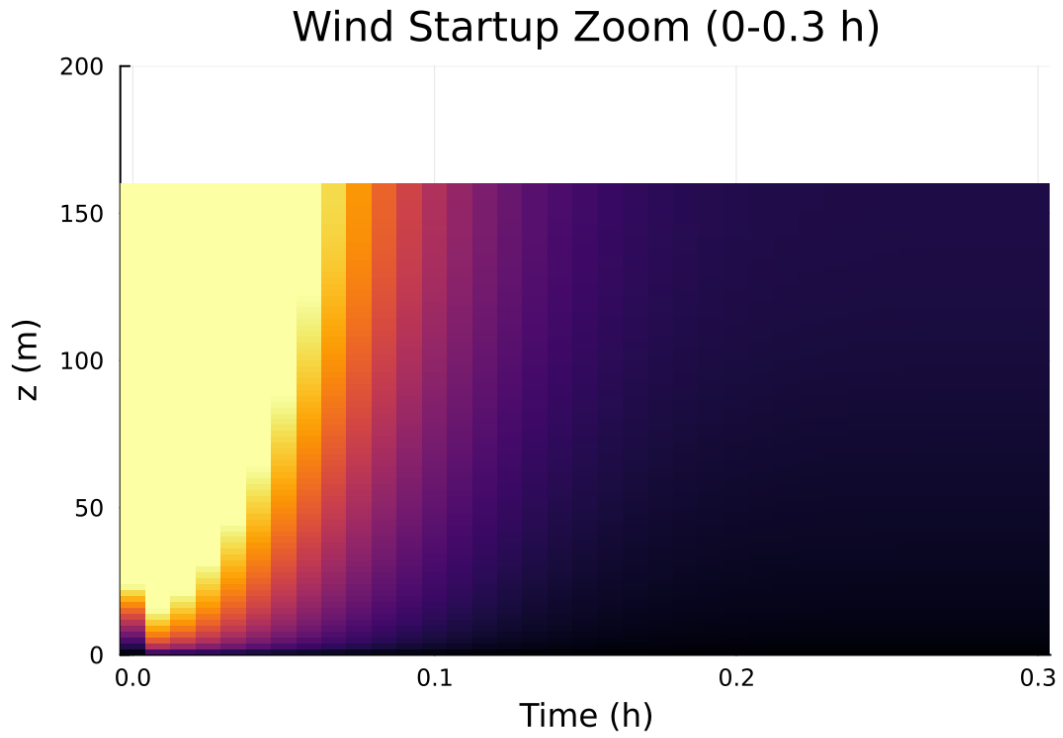


Figure 63: fig03c wind startup zoom.png

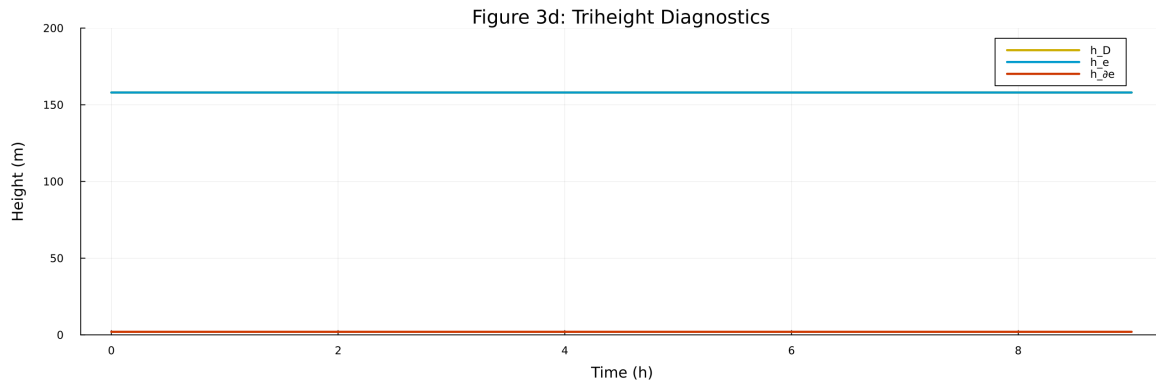


Figure 64: fig03d triheight timeseries.png

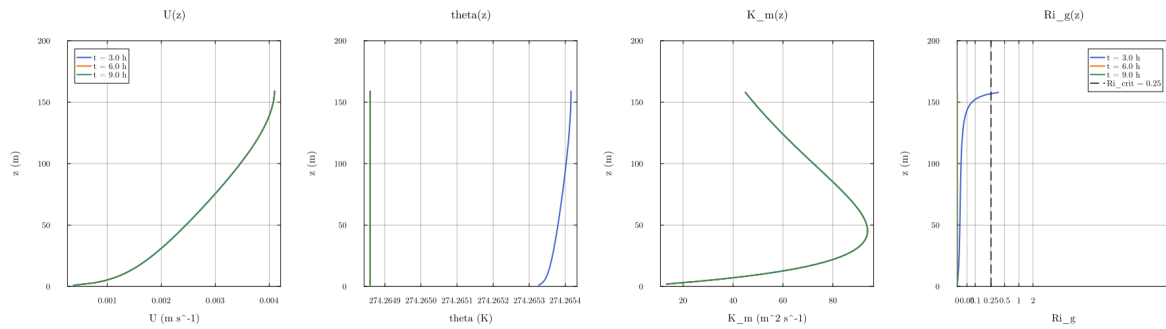


Figure 65: fig04 profiles u theta km.png

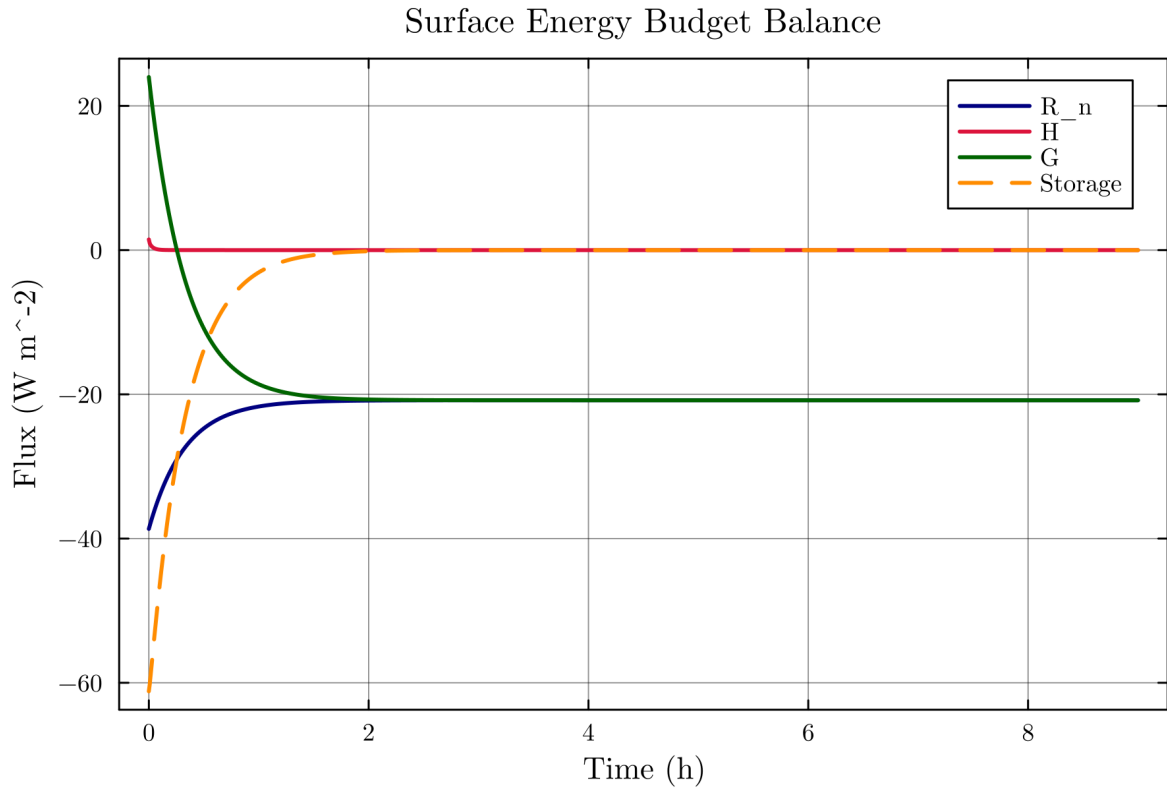


Figure 66: fig05 surface energy budget.png

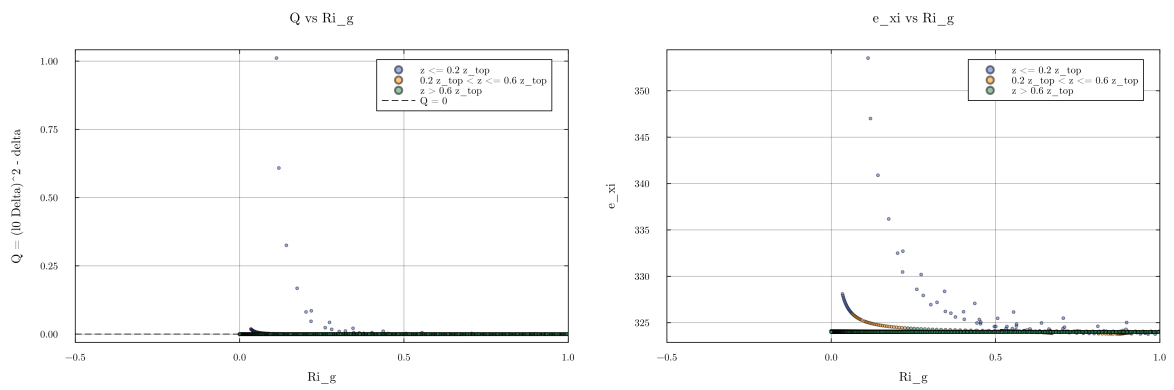


Figure 67: fig06 phase delta exi.png

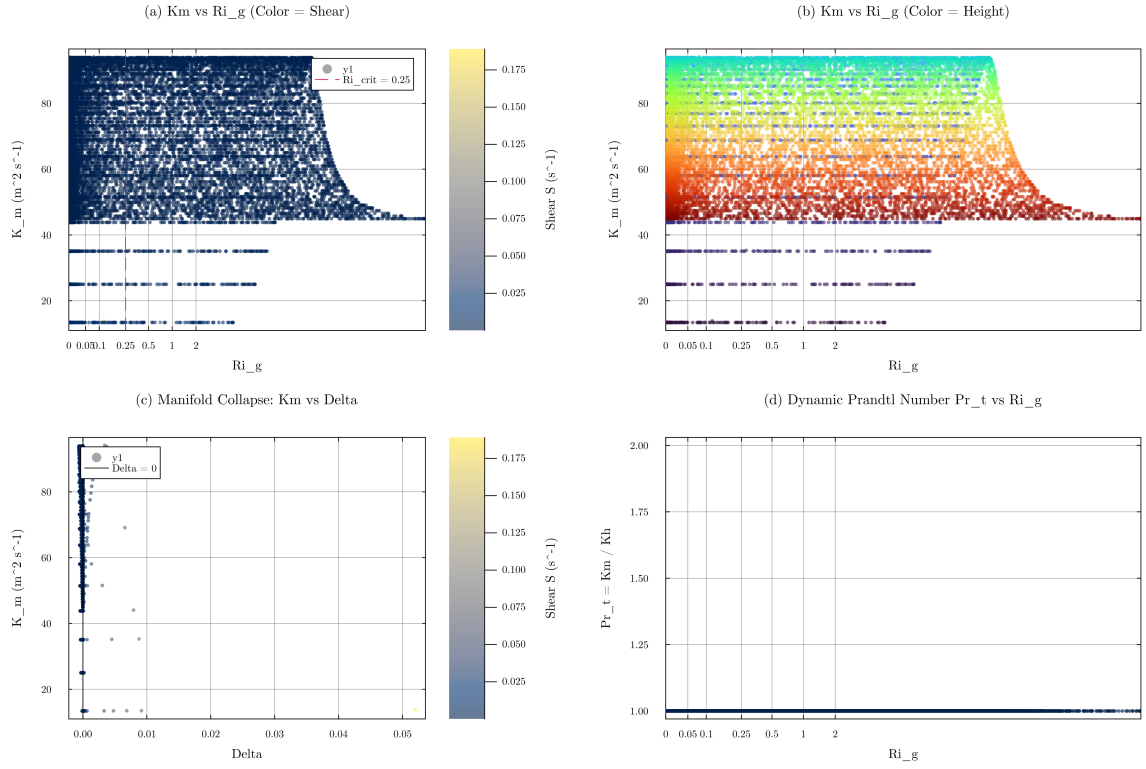


Figure 68: fig07 closure manifold.png

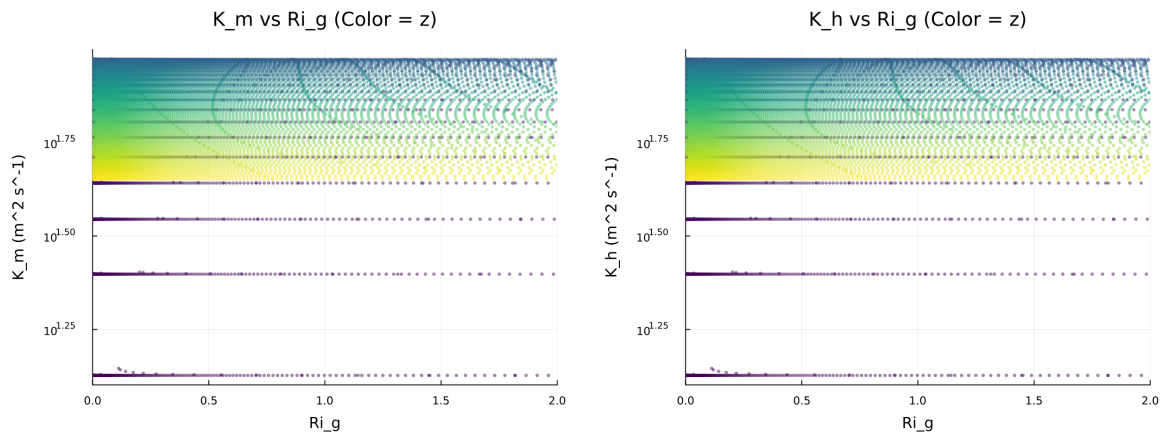


Figure 69: fig07 diffusivity vs ri.png

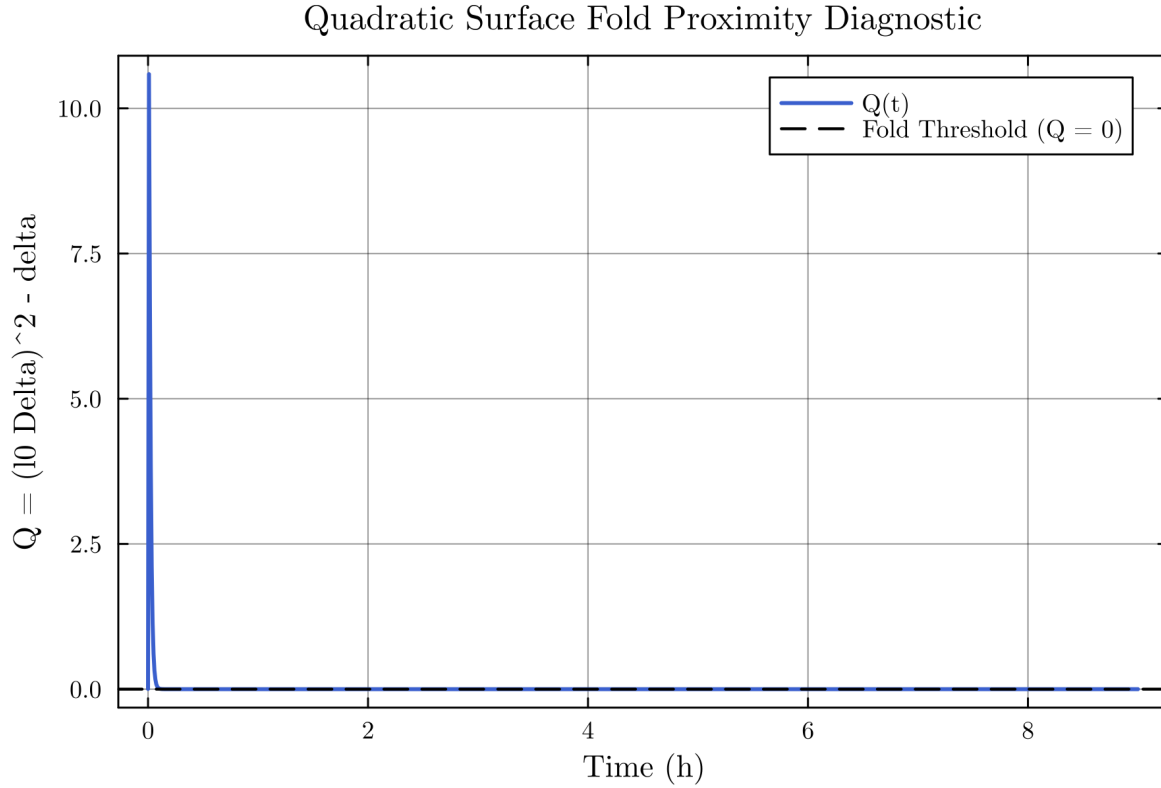


Figure 70: fig08 fold proximity.png

13.5 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

13.6 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Closure manifold diagnostics (K_m vs Ri_g and K_m vs Δ)
8. Fold proximity diagnostics versus time

13.7 Included Plots

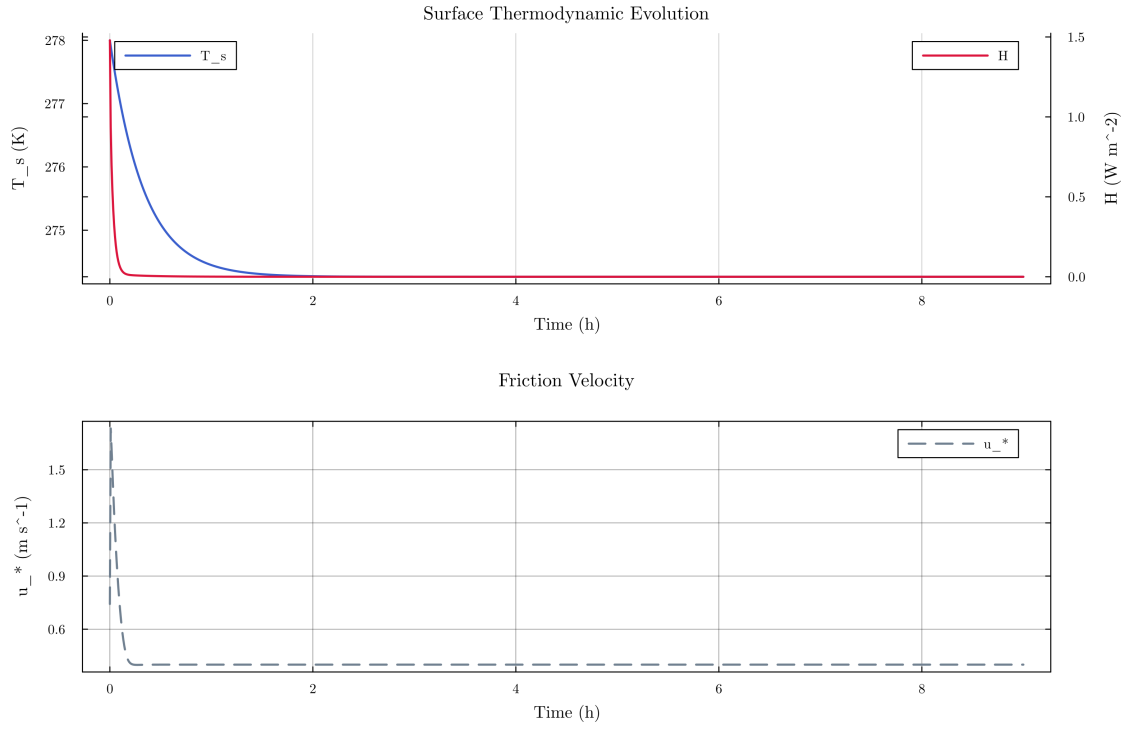


Figure 71: Time series of T_s , sensible heat flux, and friction velocity

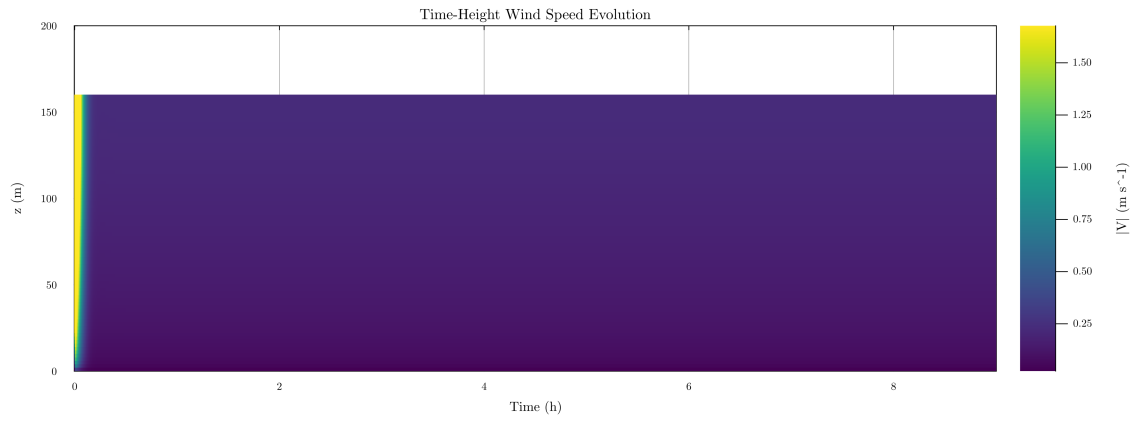


Figure 72: Time-height contour of wind speed with LLJ evolution

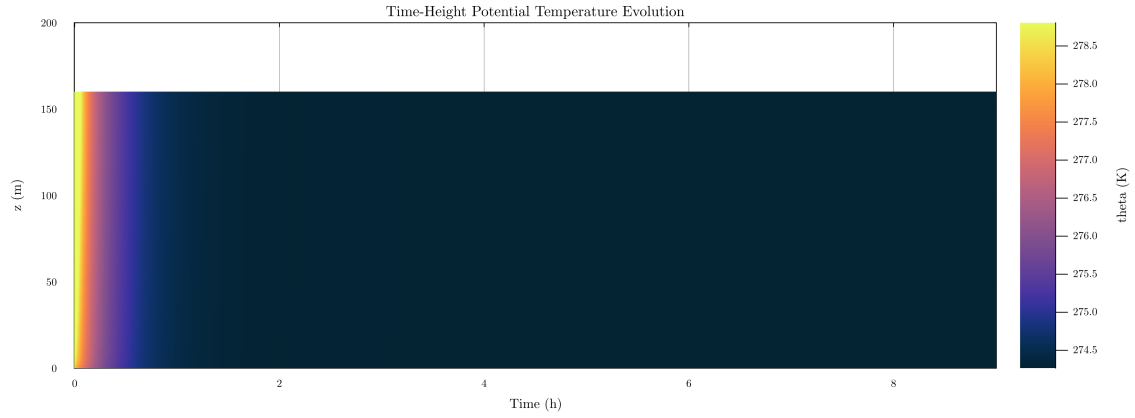


Figure 73: Time-height contour of potential temperature and inversion growth

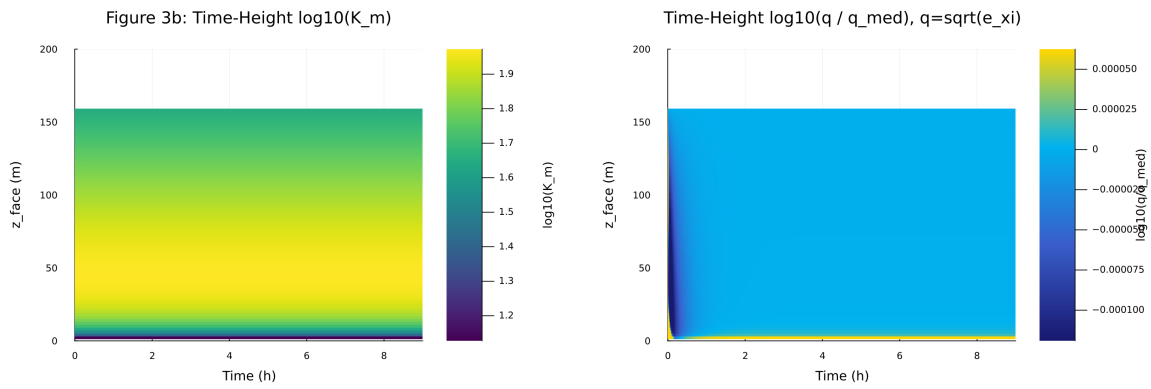


Figure 74: Vertical profiles of U , θ , and K_m at representative times

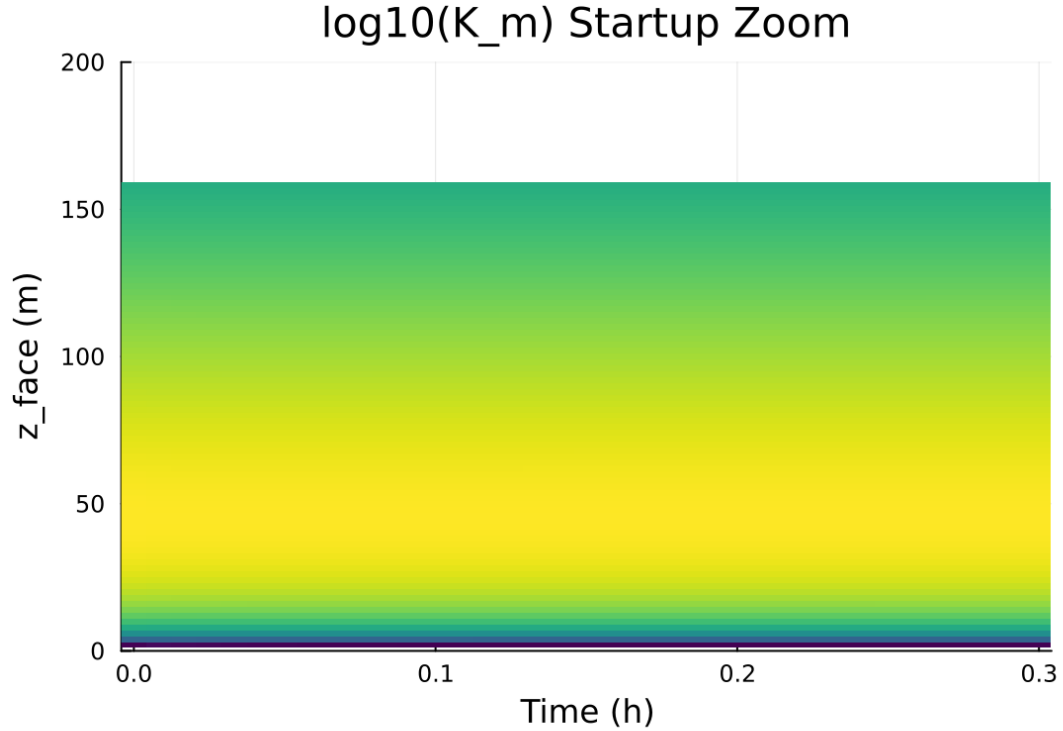


Figure 75: Surface energy budget components (R_n , H , G , storage)

14 Case 7: scm verify 40x4m 0h report

15 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/scm_verify/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

15.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/scm_verify`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 52 / 0
- RHS evaluations: 6762
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

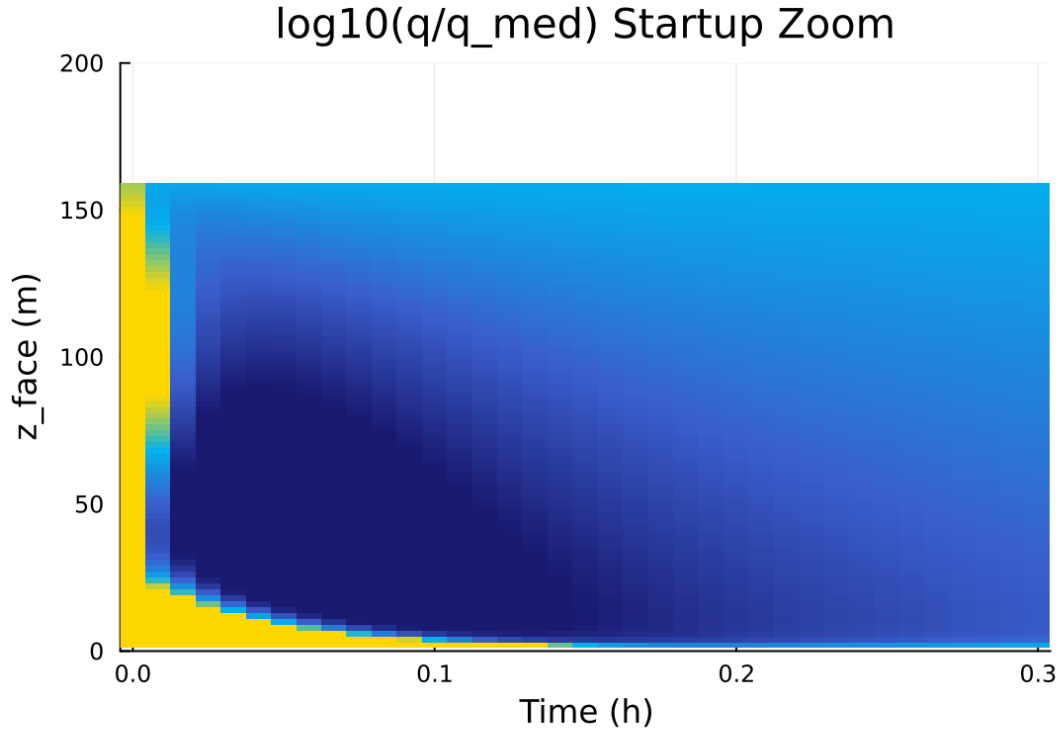


Figure 76: Phase portrait on regularized manifold (Delta vs e_{xi})

Table 13: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

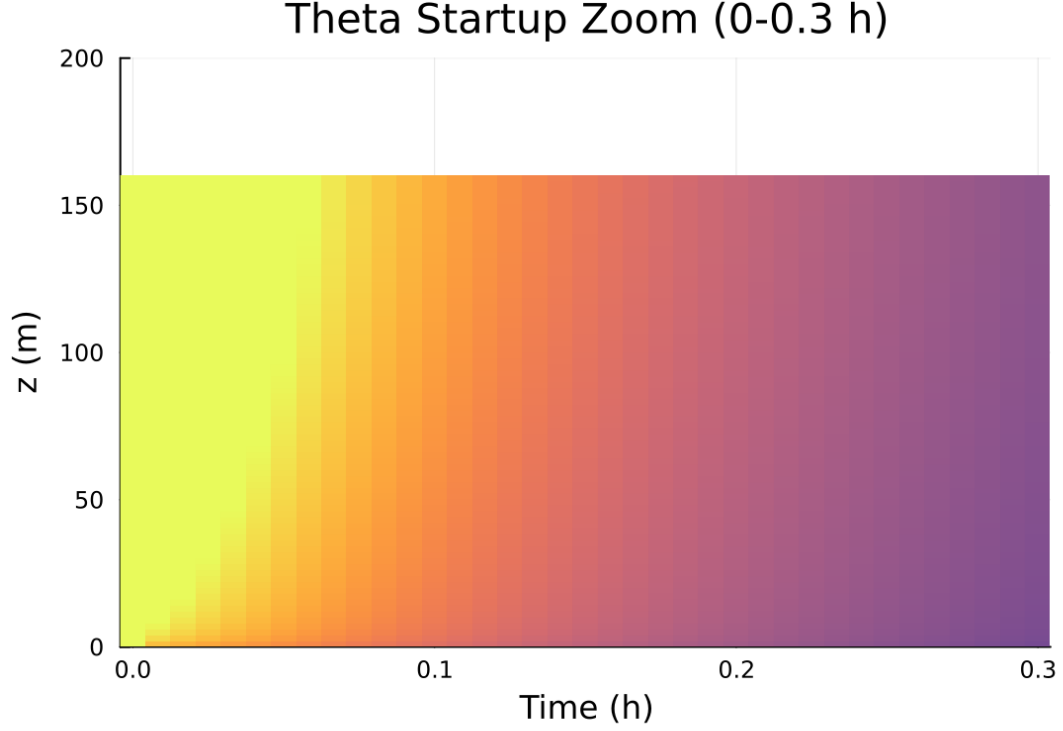


Figure 77: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)

15.2 Forcing and Boundary Conditions

15.3 GSPT Bifurcation & Manifold Calibration

Table 14: GSPT Normal Form Calibration Parameters for idealized_sbl

Metric	Symbol	Estimated Value
Report / Calibration Status	—	not_available
Estimated Fold Richardson Number	$\hat{Ri}_{\text{fold}}$	NA
95% MBB Confidence Interval	CI_{95}	[NA, NA]
Transcritical Ignition Proxy	$\hat{Ri}_{\text{critical}}$	NA
Hysteresis Width	ΔRi_H	NA
Scaling Exponent	$\hat{\gamma}$	NA
Normal Form Amplitude	$\hat{c}$	NA

15.4 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [25.002388, 94.099429] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.035577, 103.545808]$

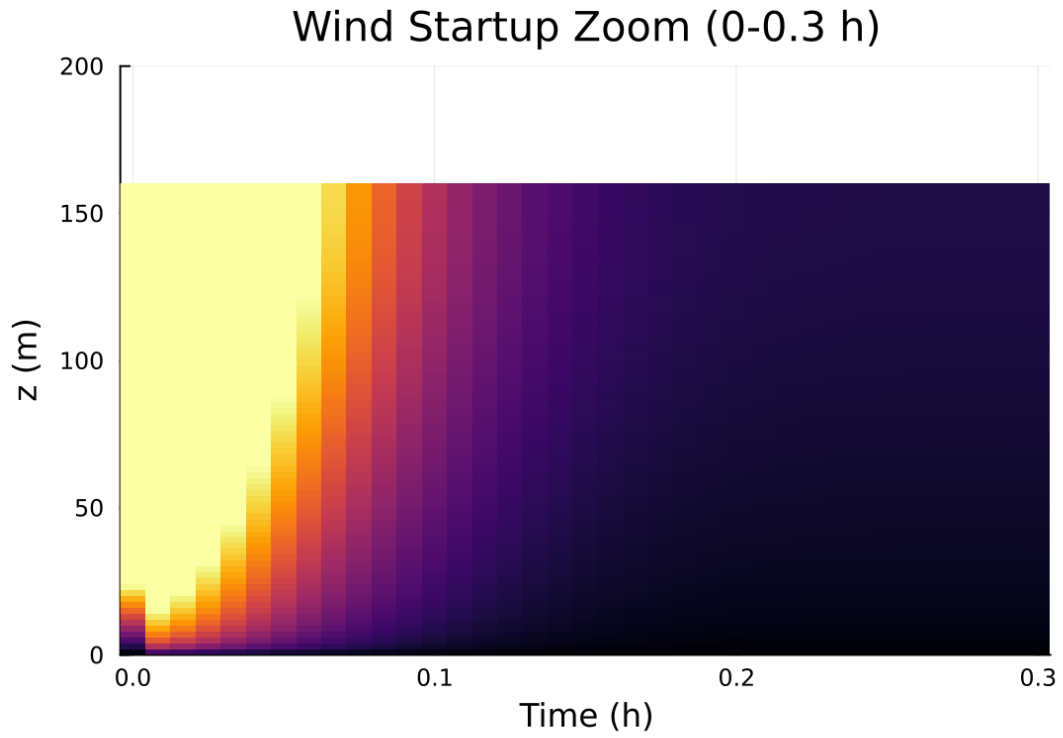


Figure 78: Fold proximity diagnostics versus time

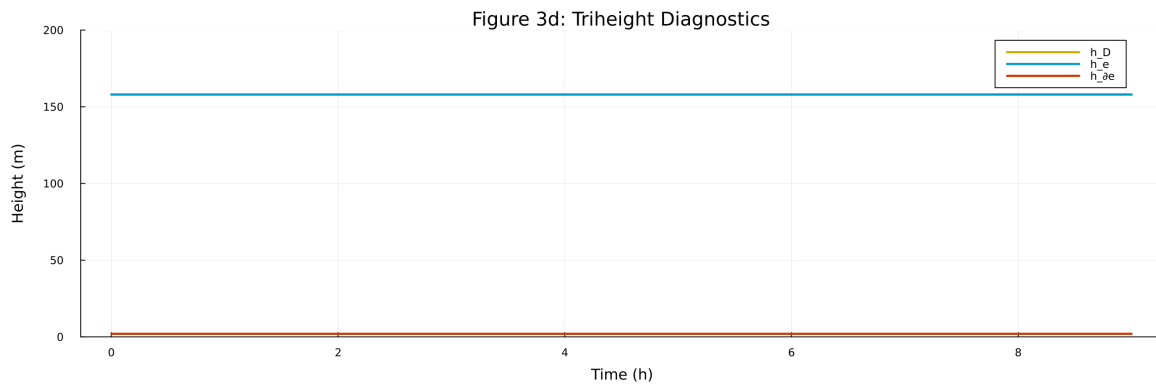


Figure 79: fig03d triheight timeseries.png

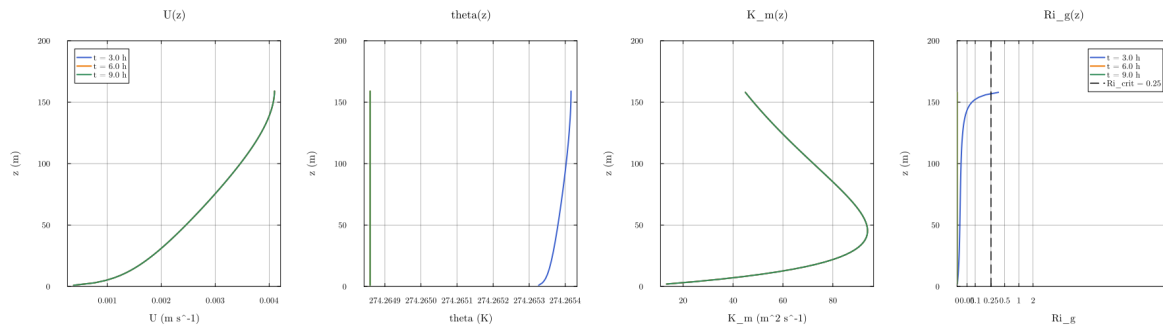


Figure 80: fig04 profiles u theta km.png

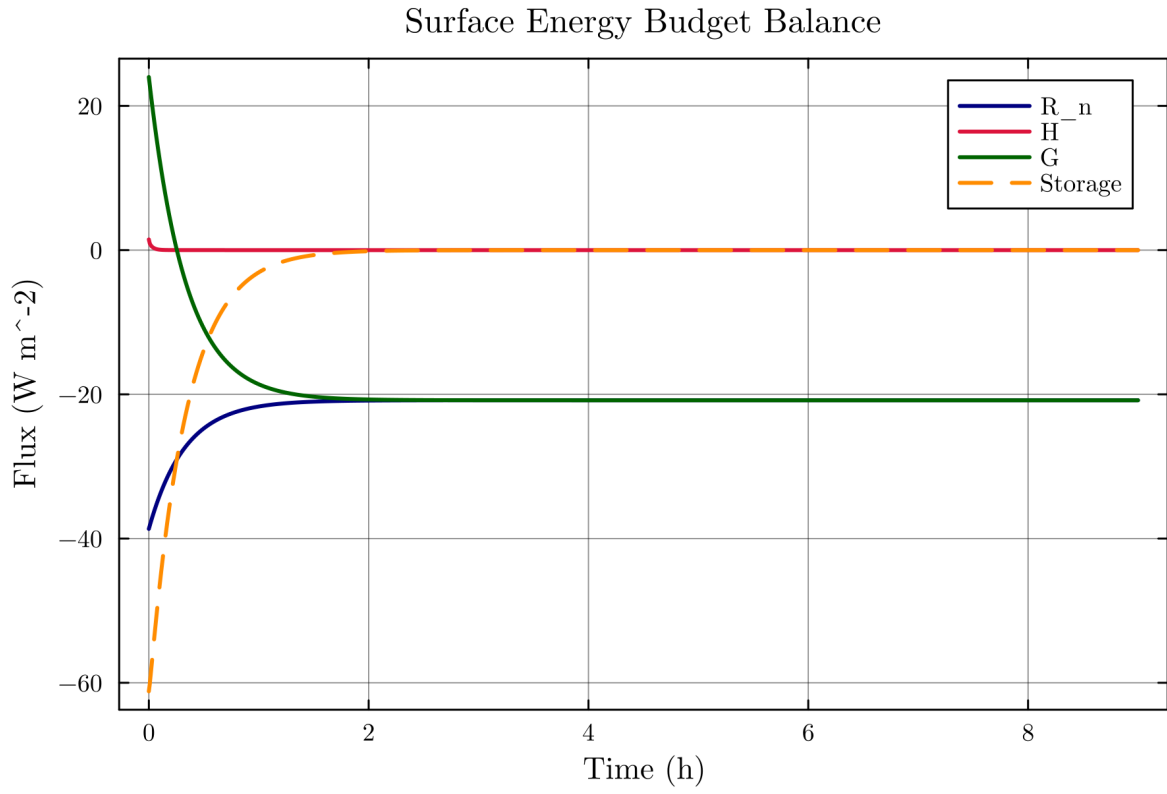


Figure 81: fig05 surface energy budget.png

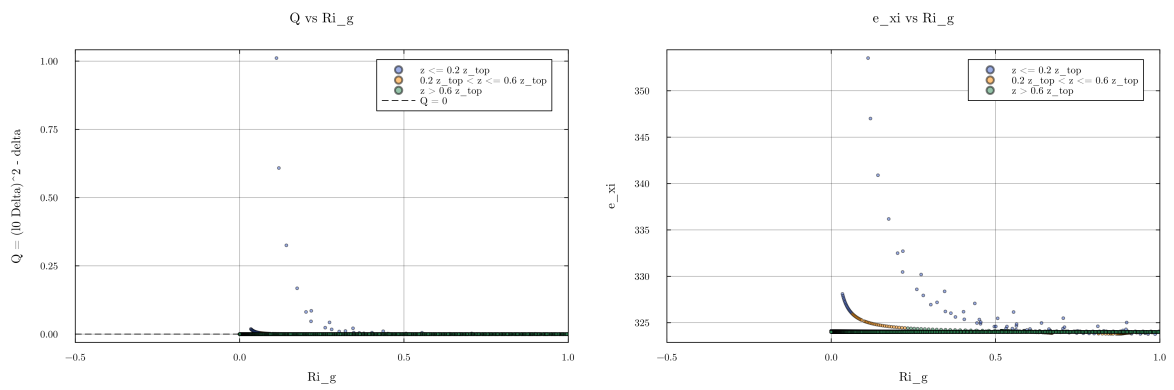


Figure 82: fig06 phase delta exi.png

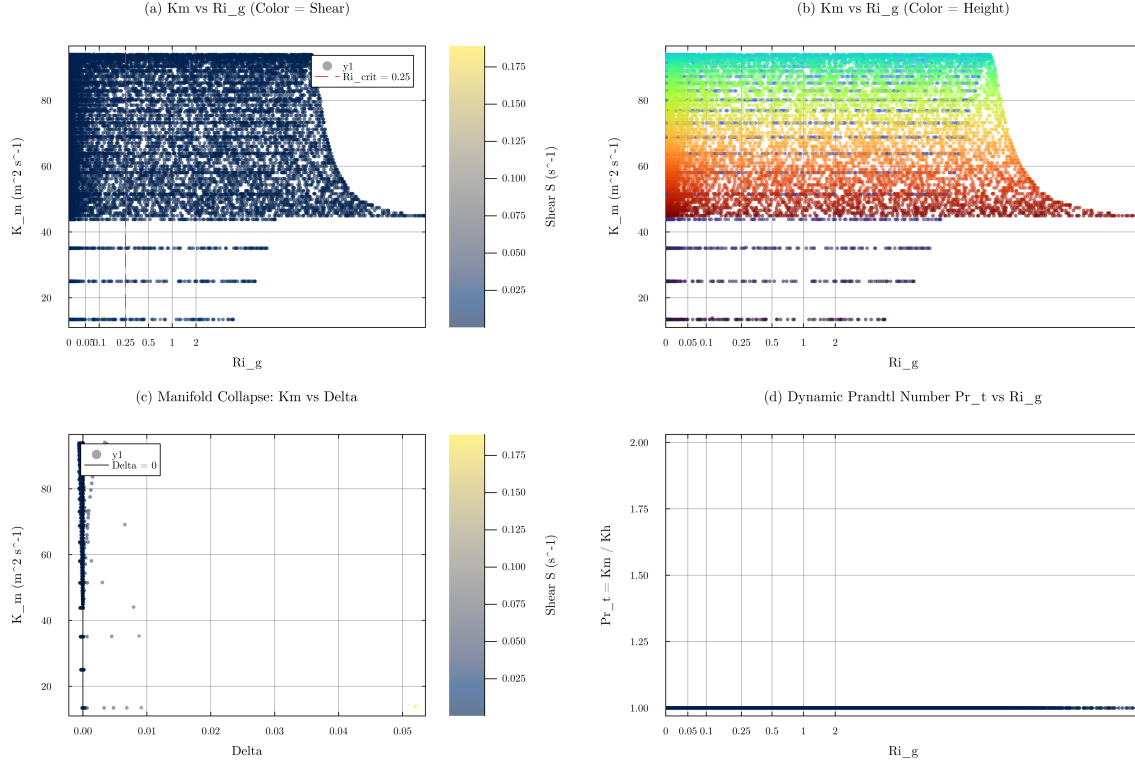


Figure 83: fig07 closure manifold.png

Quadratic Surface Fold Proximity Diagnostic

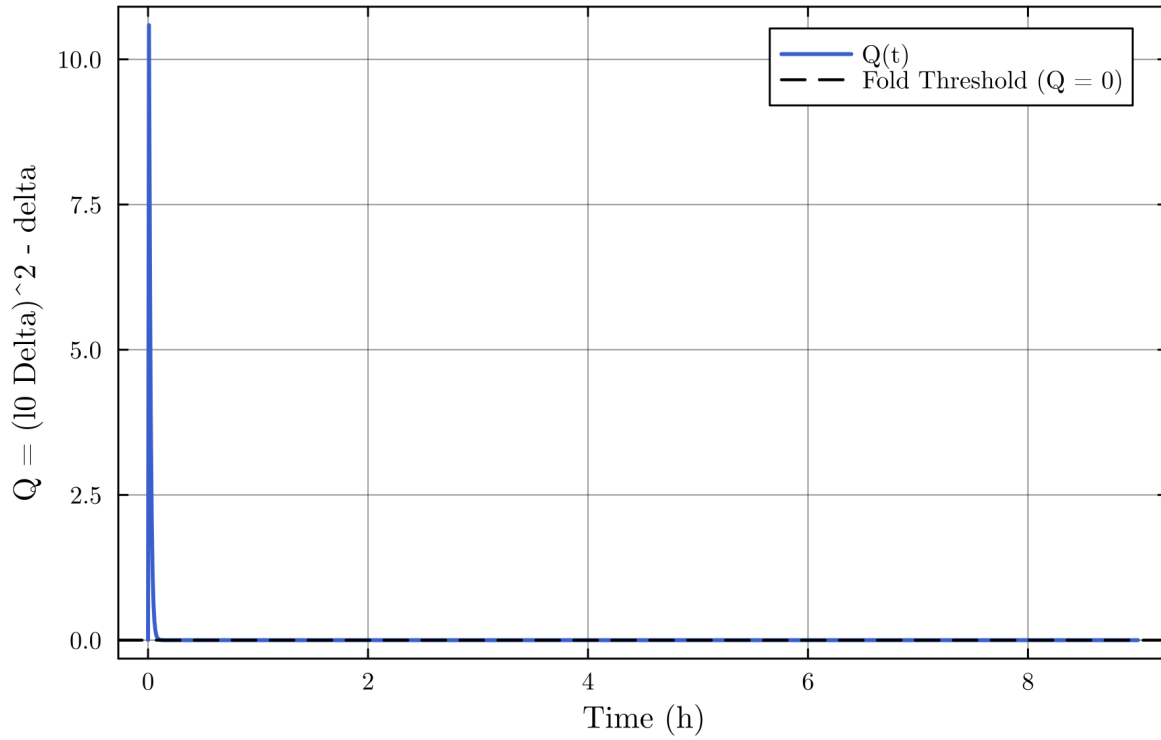


Figure 84: fig08 fold proximity.png

15.5 Generated Artifacts

- Time-series CSV: `results/scm_verify/time_series.csv`
- Diagnostic summary JSON: `results/scm_verify/summary.json`
- Payload JLD2: `results/scm_verify/payload.jld2`

15.6 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Closure manifold diagnostics (K_m vs Ri_g and K_m vs Δ)
8. Fold proximity diagnostics versus time

15.7 Included Plots

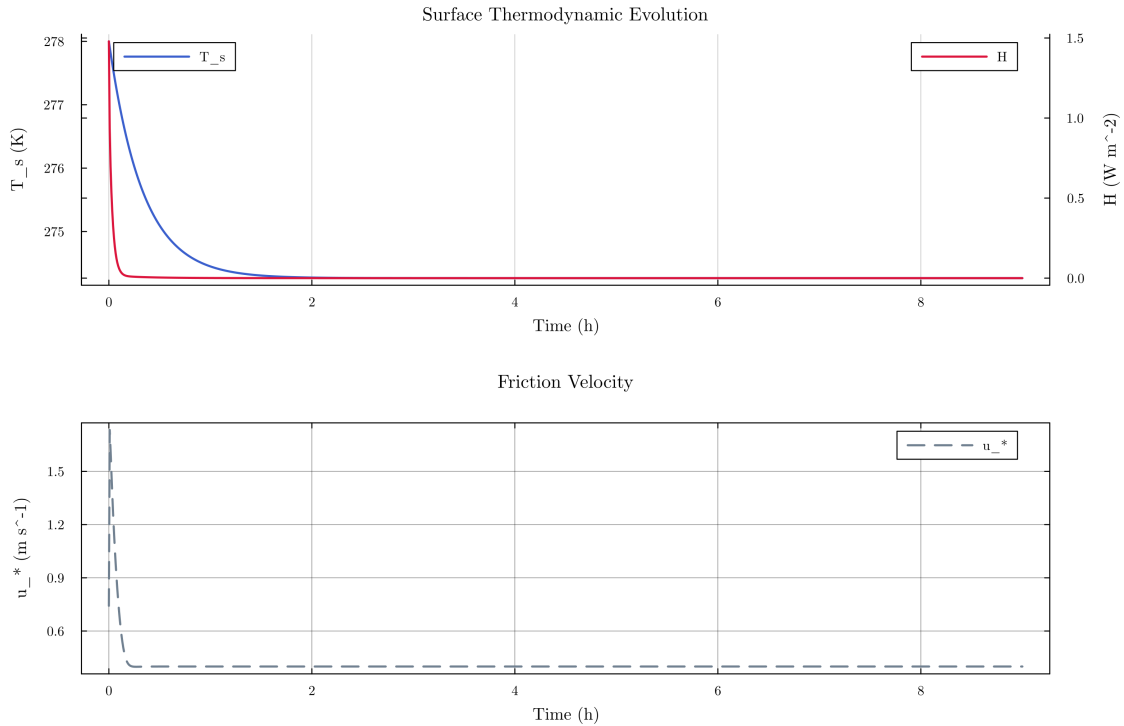


Figure 85: Time series of T_s , sensible heat flux, and friction velocity

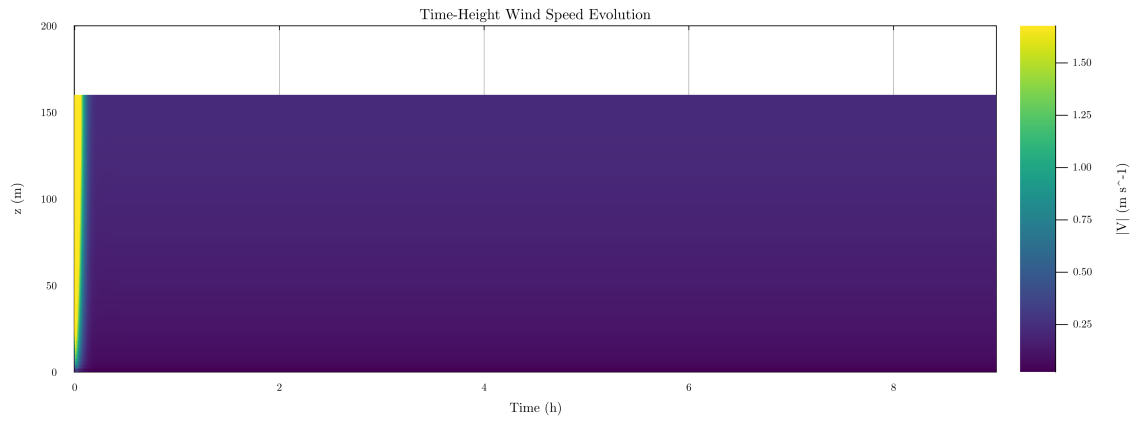


Figure 86: Time-height contour of wind speed with LLJ evolution

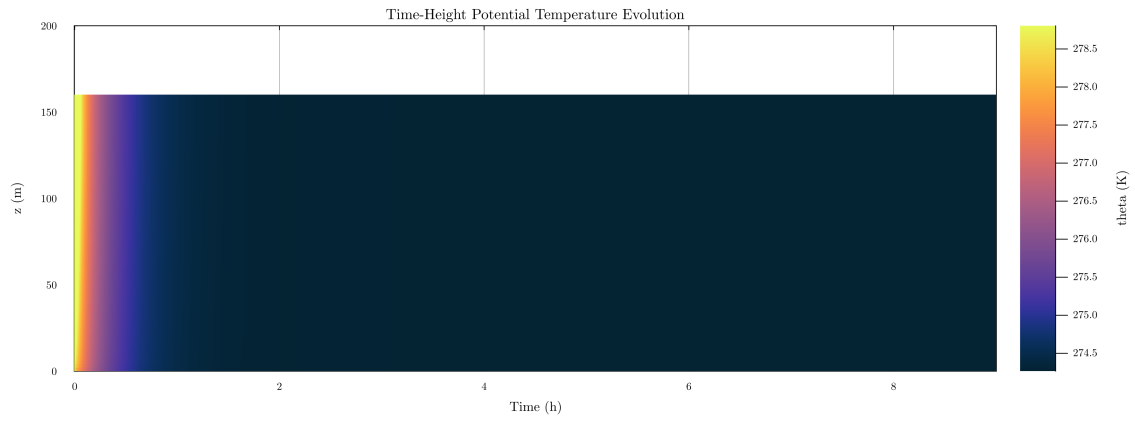


Figure 87: Time-height contour of potential temperature and inversion growth

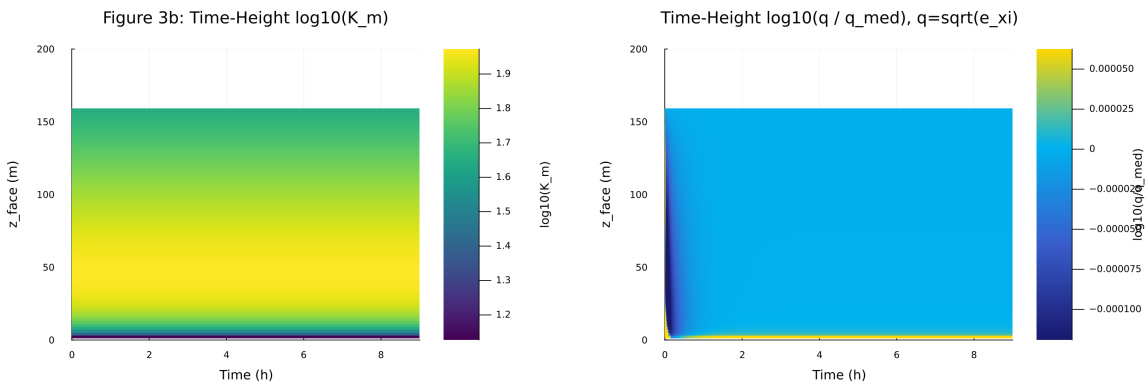


Figure 88: Vertical profiles of U , θ , and K_m at representative times

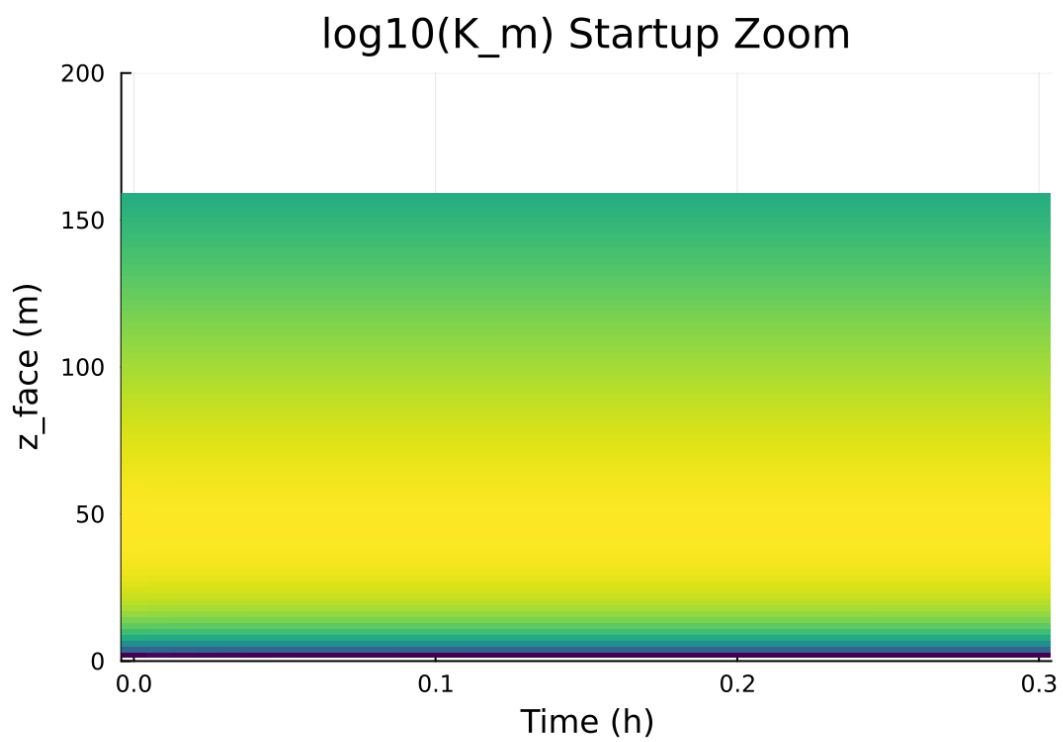


Figure 89: Surface energy budget components (R_n , H , G , storage)

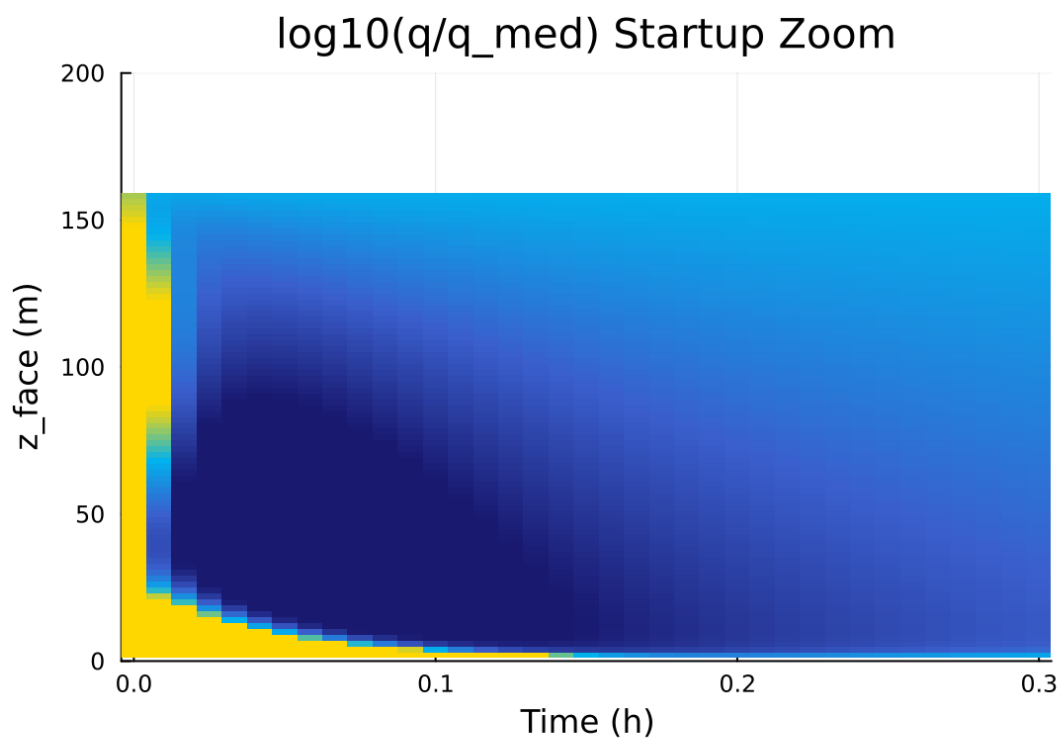


Figure 90: Phase portrait on regularized manifold (Delta vs e_{xi})

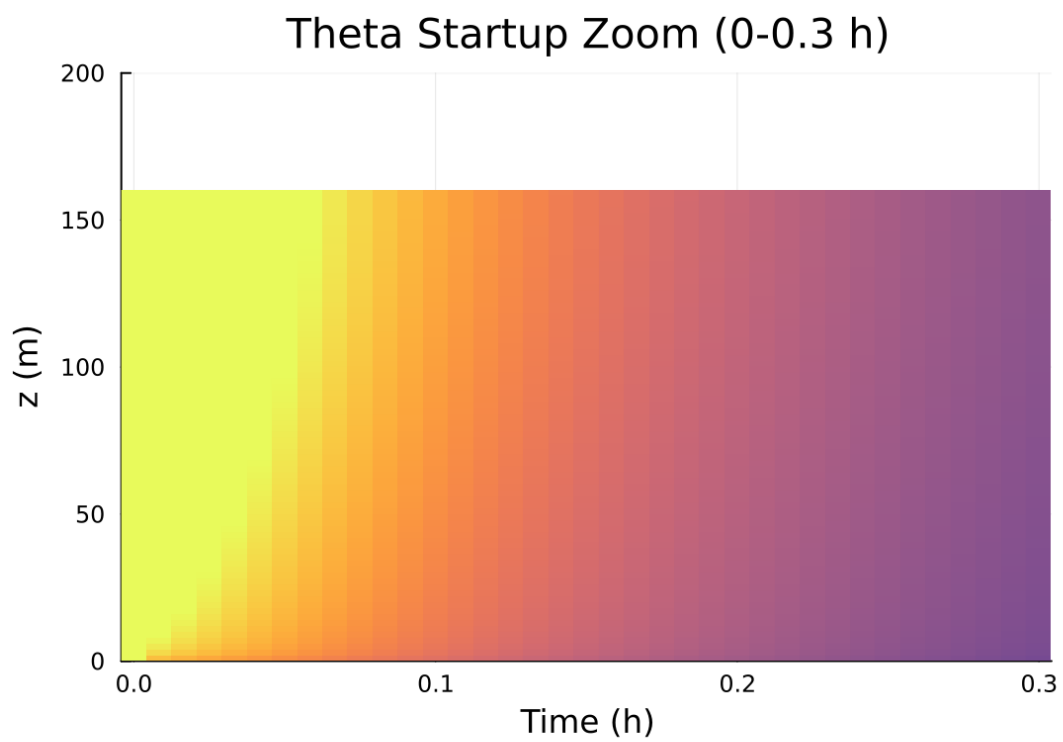
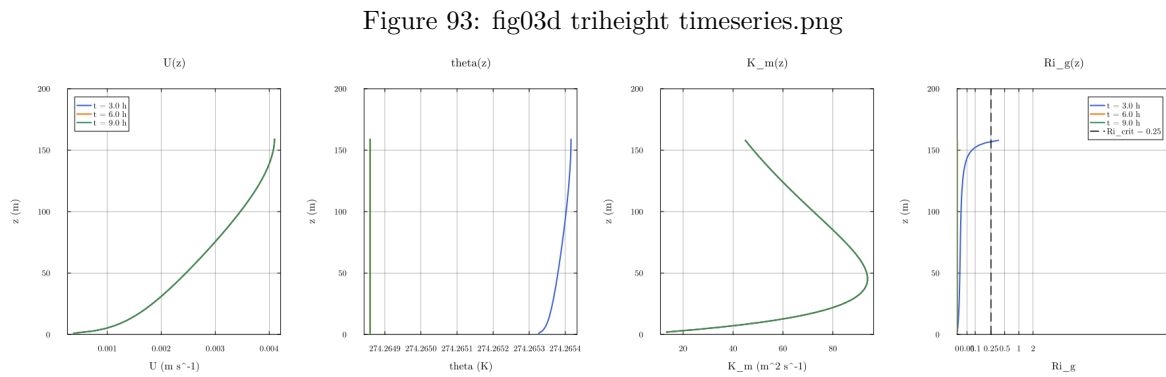
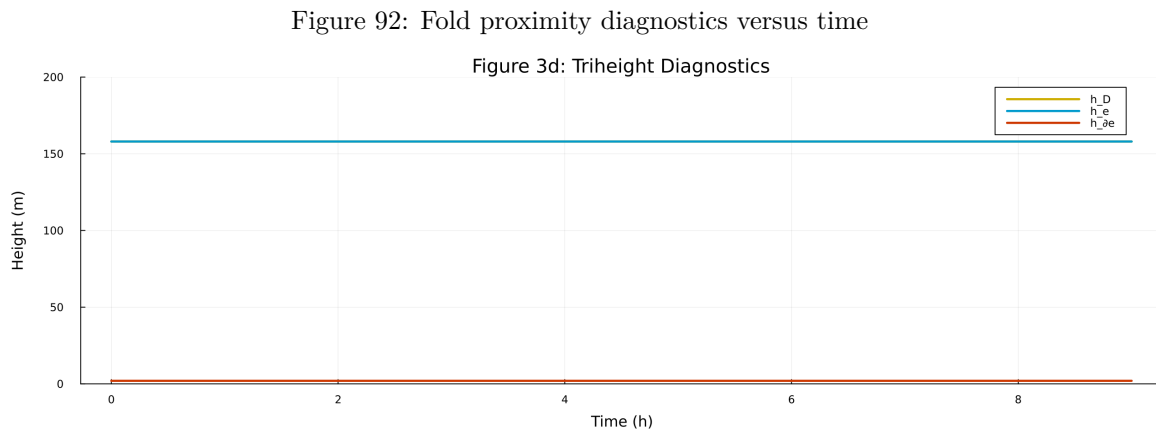
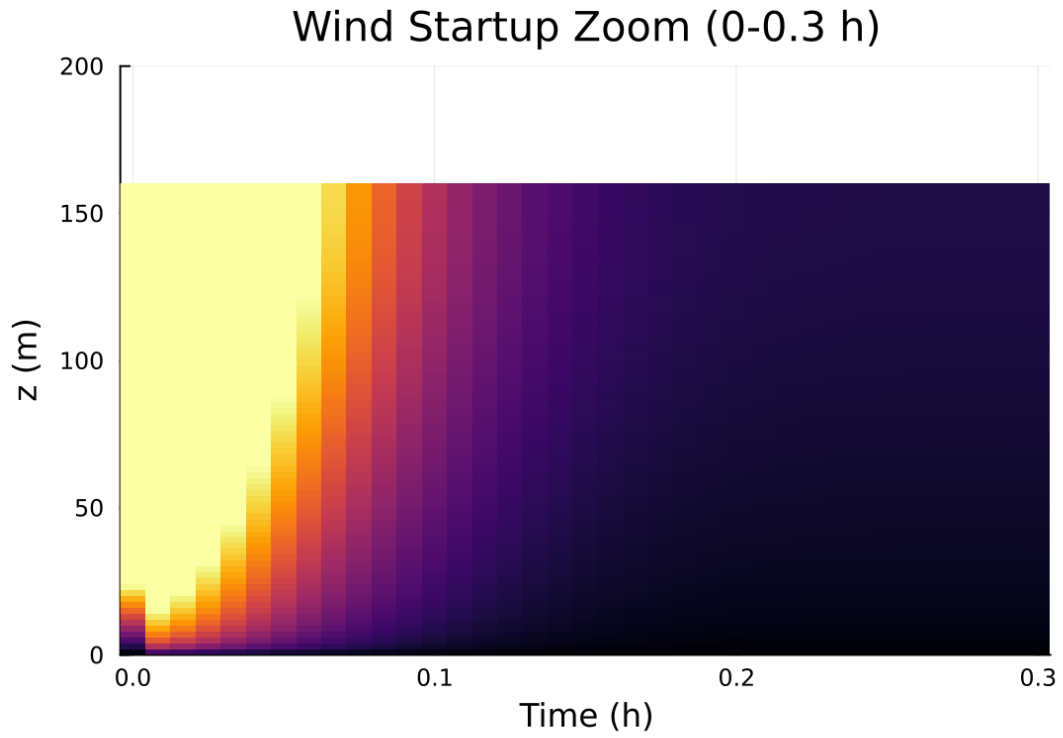


Figure 91: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)



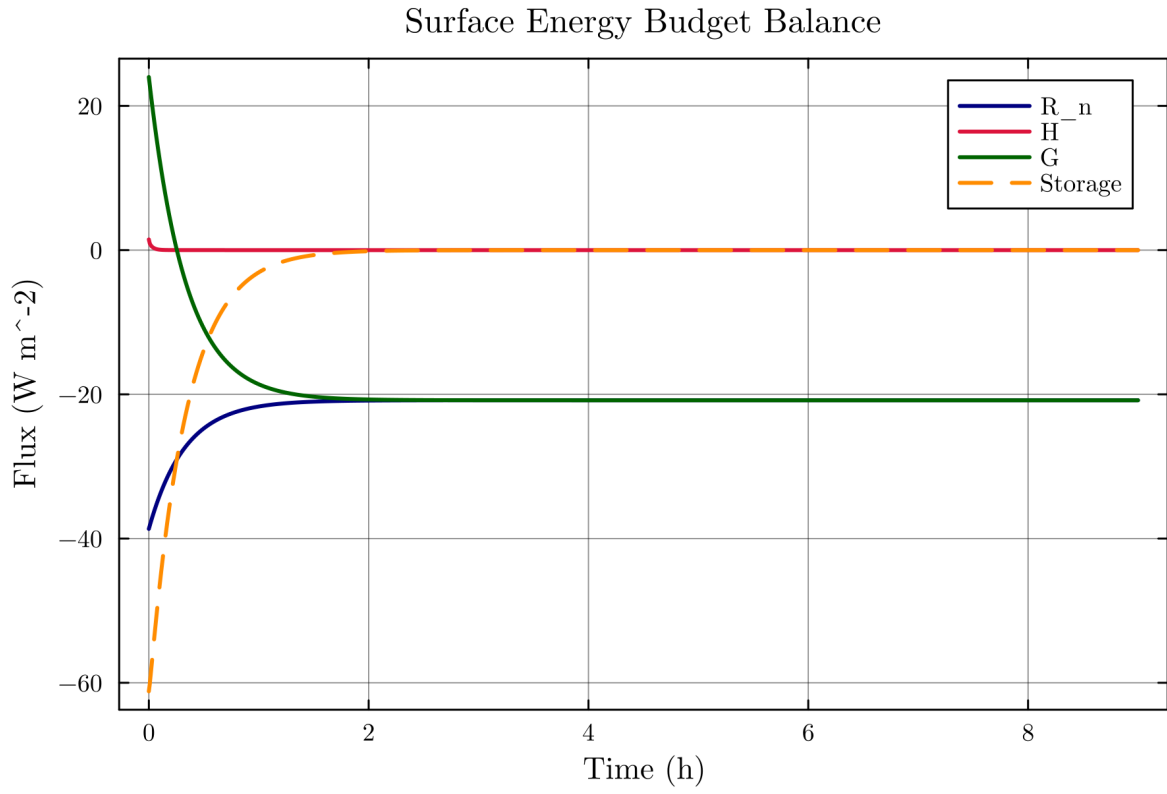


Figure 95: fig05 surface energy budget.png

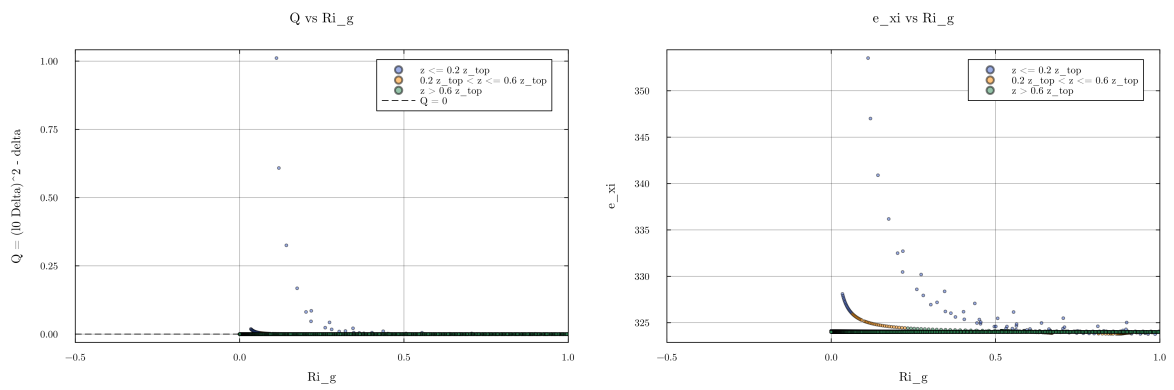


Figure 96: fig06 phase delta exi.png

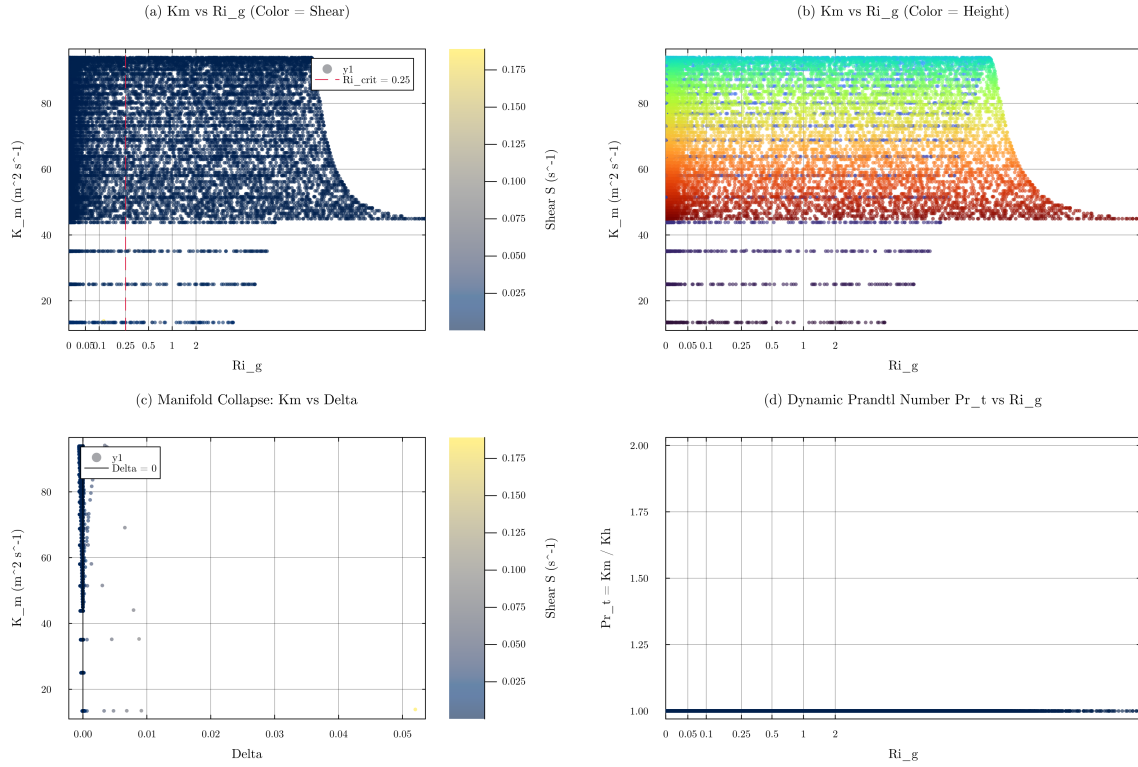


Figure 97: fig07 closure manifold.png

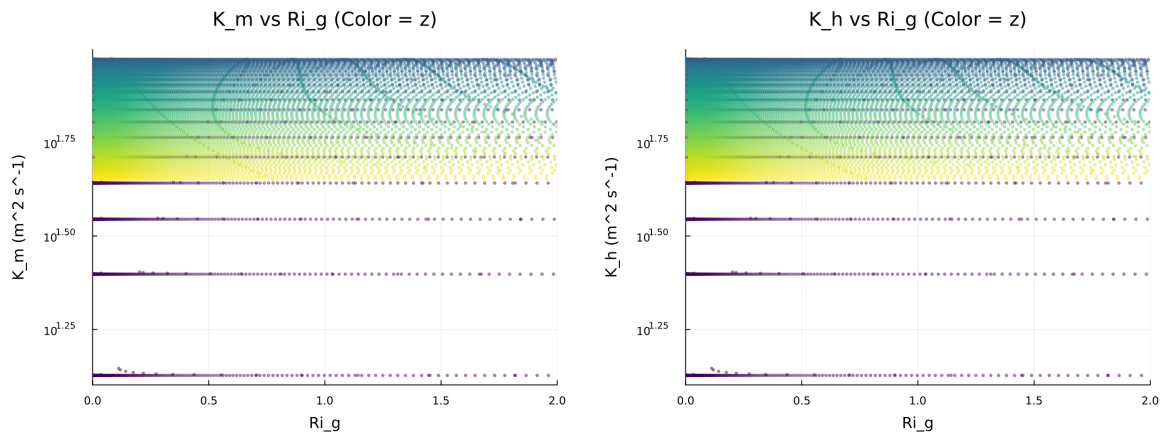


Figure 98: fig07 diffusivity vs ri.png

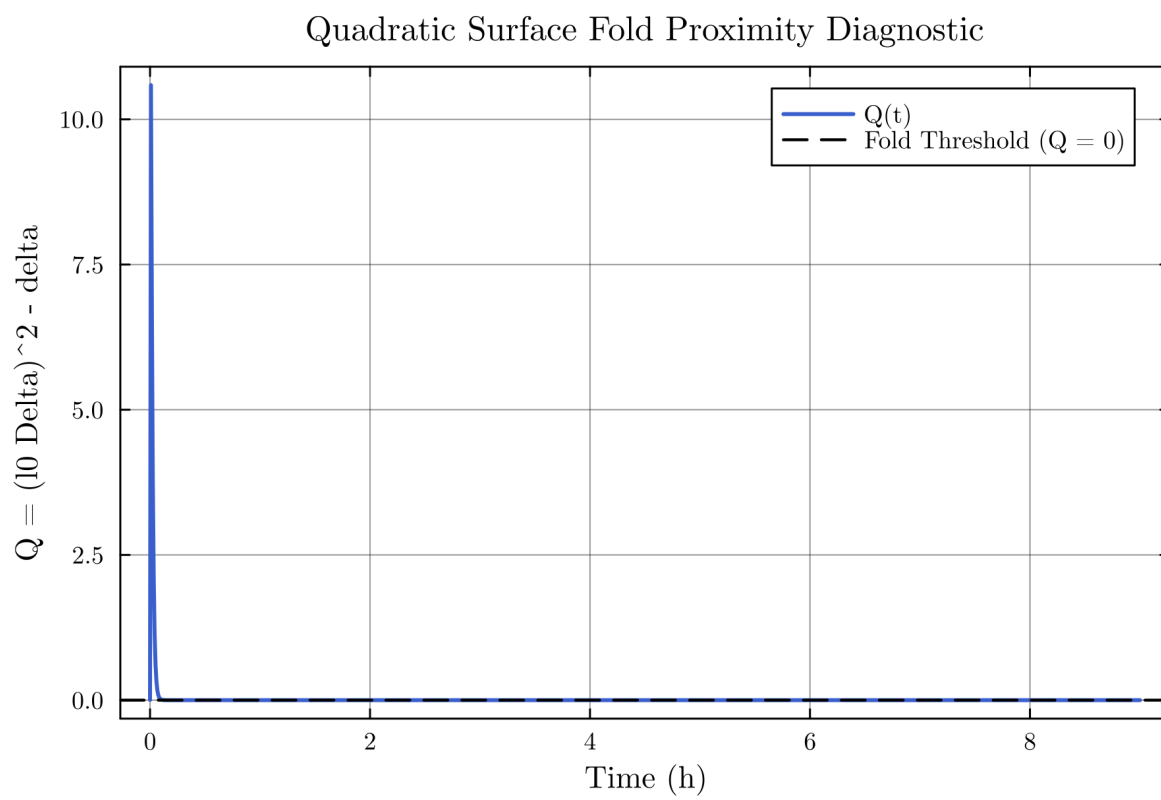


Figure 99: fig08 fold proximity.png